

Modern Eddington

Experiment 2024: Results and Conclusions

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ABSTRACT

The main goal of the Modern Eddington Experiment 2024 (MEE2024) was to acquire images of stars near the disk of the Sun during the total eclipse of April 8th, 2024. By precisely measuring the gravitational shift in the stars' apparent locations, the deflection coefficient $L = 4GM_{\odot}/R_{\odot}c^2$ predicted by Einstein's theory of general relativity could be measured. The experiment was attempted by a total of thirteen telescope stations at three locations across the central path of the eclipse. While seven stations in Texas were completely clouded out, one of the stations in central Mexico had modest success, capturing 171 stars during the 4.5 minutes of totality. Our estimate for the deflection coefficient using a subset of the data from this station is $L = 1.84 \pm 0.24$ arcsec, which is consistent with Einstein's prediction of 1.7512 arcsec. This shows that with adequate planning and practice, teams of faculty, undergraduate students, and amateur astronomers can measure the deflection coefficient using amateur-grade astronomical equipment. In this paper, we present a summary of the data collected during the eclipse and a detailed description of the open source data analysis software package that was written specifically for this project. We also discuss some of the lessons that we learned while conducting the experiment and offer equipment and procedural suggestions for those attempting to repeat the experiment during the eclipses of August 2nd, 2027 and July 22nd, 2028, which should provide some of the best opportunities to measure L in the next 100+ years.

1. Introduction

In his early explorations of the consequences of the still nascent theory of general relativity, Einstein (1911) predicted that the path of a beam of light traveling past a massive body would deviate from a straight line path. Despite an early misstep in 1911 that led him to miss a factor of 2, he would eventually arrive at the now accepted expression for the angle by which the light ray would be deflected as a function of its distance from the disk of the Sun (Einstein, 1916). Specifically, Einstein concluded that a ray of light passing near the limb of the Sun would be gravitationally deflected by an angle

$$\Delta\theta = \frac{4GM_{\odot}}{R_{\odot}c^2} \frac{1}{R}, \quad (1)$$

where G is the universal gravitational constant, $M_{\odot}$ and $R_{\odot}$ are the solar mass and radius, c is the speed of light in vacuum, and R is the distance at which the light ray passes from the center of the Sun measured in solar radii. Because of this, when a star is close to the disk of the Sun in the sky, its apparent angular position will change by $\Delta\theta$ as shown in Figure 1.

The combination of parameters $L = 4GM_{\odot}/R_{\odot}c^2$ is called the Einstein deflection coefficient and has a numerical value of 1.7512 ± 0.0003 arcsec.¹ Besides being a very small effect, measuring the bending of starlight around the Sun poses the particular challenge that, under most conditions, stars that are near the Sun in the sky are too dim to observe over the Sun's glare. Therefore, scientists had to wait until a total solar eclipse,

when the disk of the Sun would be blocked by the Moon and stars would become visible, to look for evidence of the effect.

Early attempts to test Einstein's predictions ([Campbell & Curtis, 1914](#); [Perrine, 1923](#)) were thwarted by weather and the onset of World War I and it was not until the total eclipse of May, 29th, 1919 that Einstein's prediction could be put to the test ([Earman & Glymour, 1980](#)). On that day, twin expeditions organized by Sir Frank Dyson and Sir Arthur Eddington to the West African island of Príncipe and the town of Sobral in northern Brazil were able to capture a handful of plates. From the two plates captured in Sobral, they were able to measure the apparent change in position for seven stars. This deflection was found to be consistent with Einstein's predictions ([Dyson et al., 1920](#)) and determined to be definitive evidence in support of Einstein's theory, immediately making him a household name ("[Revolution in science](#)", 1919; "[Light all askew](#)", 1919).²

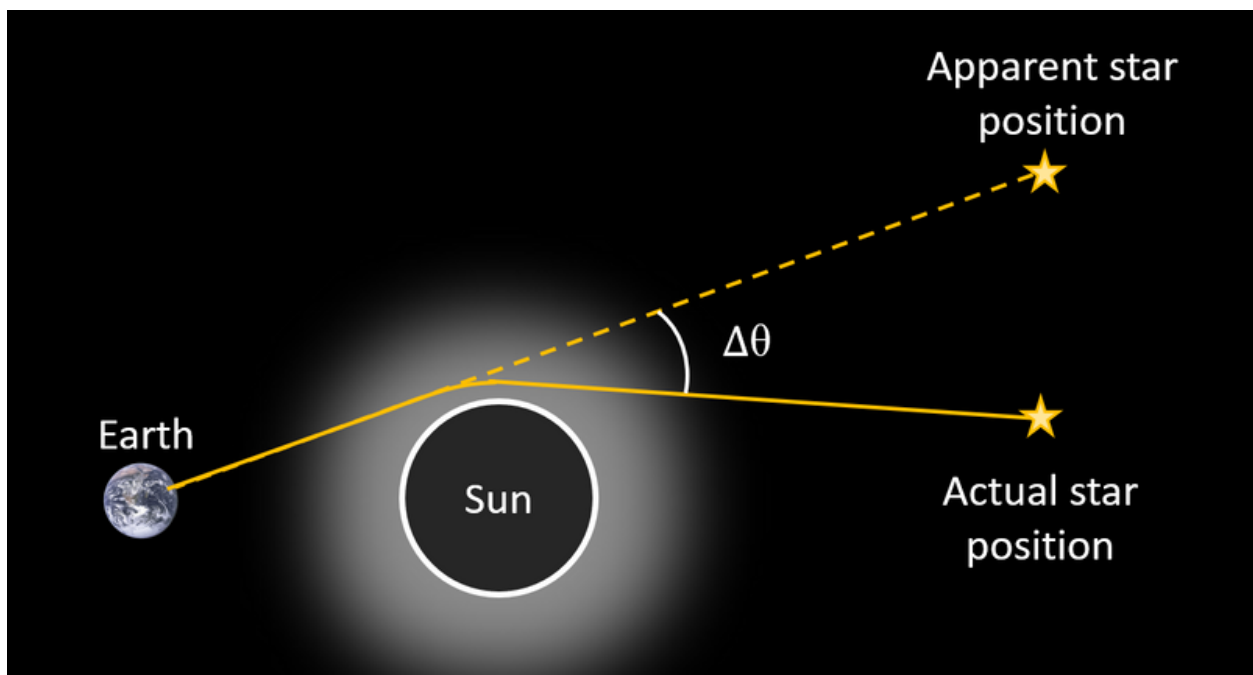


Figure 1

Schematic representation of the bending of a beam of light as it passes near the disk of the Sun. Because of this deflection, the apparent position of the star in the sky appears to change by an angle $\Delta\theta$. Note that actual deflection angles are on the order of arcseconds or smaller and *much* smaller than the angle shown here. For reference, an arcsecond is about the angular size of a US dime observed from 4 km away.

In the decades following what came to be called the Eddington experiment, a number of expeditions were launched to reproduce and improve on the original result (see von Klüber ([1960](#)) and Will ([2015](#)) for comprehensive reviews of these attempts). Following a moderately successful attempt by a team from the University of Texas in 1973 ([Brune Jr. et al., 1976](#)), there was (to the best of our knowledge) a 44 year gap with no new attempts at measuring the deflection coefficient using ground-based optical telescopes until the Great

American Eclipse of 2017.³ In that year, experiments using modern digital cameras and star catalogs were conducted by several parties, ushering in the era of the Modern Eddington Experiment (MEE).

Three groups were successful in conducting MEE experiments in 2017 – Bruns at a 7840 ft. elevation on Casper Mountain in Wyoming ([Bruns, 2018](#)), Kinne (set up near Bruns on Casper Mountain), and a team of four undergraduate students at the Alpaca Meadows Observatory near Lyons, Oregon supervised by Berry and Dittrich. Each of these teams used small refactor telescopes and CCD (charge-coupled device) cameras to acquire ~ 40 images of a few dozen stars near the disk of the Sun. Bruns obtained a deflection coefficient 1.752 arcsec with an uncertainty of 3%, making his measurement the most accurate and precise determination of L using ground-based optical telescopes in history ([Grant, 2018](#)). Unpublished data sets from Kinne and from Berry and Dittrich’s team gave values of 2.03 arcsec and 1.68 arcsec, respectively, although with significant uncertainties. Importantly, the experiment performed by students under the guidance of Berry and Dittrich demonstrated that this challenging experiment could be performed by undergraduate students ([Berry, 2022](#)).

These attempts gave impetus for the creation of teams to repeat the experiment during the total eclipse of 2024. Leading up to the eclipse, a paper was published ([Dittrich et al., 2024](#)) outlining the goals and methods of the experiment, which was given the name MEE2024 and several teams were put together to attempt observations at multiple locations along the eclipse path. This paper reports the initial results from these observations. Lessons learned and suggestions for optimizing the equipment and procedures for the total eclipses of 2027 and 2028 are also presented. As will be discussed in Section 5.4, these eclipses provide some of the best opportunities for high precision Eddington experiments that will be available in the next 100+ years.

2. The Modern Eddington Experiment of 2024

Following the recommendations of Bruns ([2016, 2018](#)), Berry ([2022](#)) and Dittrich et al. ([2024](#)), preparations were undertaken to conduct the Modern Eddington Experiment during the eclipse of April 8th, 2024. This section discusses the makeup of the teams and the equipment they used, the locations of the sites selected, and discusses complications encountered due to unexpected weather conditions on the day of the eclipse. It closes by summarizing the overall goals of the MEE2024 project.

2.1 - April 8th, 2024 Observation Sites

In preparation for the eclipse, thirteen telescope/camera stations were organized and distributed among three sites (see [Figure 2](#)). These were led by faculty from several universities, staff from public observatories, and amateur astronomers. The stations were operated by students who trained with the equipment during the months leading up to the eclipse. The eclipse first passed over a team with 14 participants led by W. A. Dittrich and D. Borrero-Echeverry located at an elevation of 7927 ft at a site east of the town of El Salto (23°50'58.4" N 105°16'22.0" W) in the state of Durango, Mexico. For an account of eclipse day at the site at El Salto and the

events leading up to it, see Patel (2024). The shadow then arrived at a single telescope station operated by J. M. Izen at a location near Ciudad Canatlán, Durango, Mexico. Finally, the eclipse reached the third location in Leakey, Texas where 25 students supervised by G. Mulder, H. Hill, and K. Carrell operated 7 stations. Several additional independent amateur astronomers were also present at the Texas site.



Figure 2

(Left) MEE2024 consisted of teams at three observation sites near the center line (blue) of the path of totality (shaded). Two sites were in the Mexican state of Durango, near the towns of El Salto and Ciudad Canatlán. A third site was located outside of Leakey, Texas. (Right) The site at El Salto consisted of five stations, which were set up in a courtyard surrounded by brick walls. This turned out to be fortuitous as the walls protected the telescopes from high winds that were present in the area during the days leading up to the eclipse and which disabled the station at Ciudad Canatlán on the day of the eclipse. The map in the left hand panel was created using Google maps. The eclipse path was adapted from publicly available data posted on NASA's Goddard Space Flight Center's eclipse website, <https://eclipse.gsfc.nasa.gov>. The photograph on the right is courtesy of J. Delich.

Unfortunately, on April 8th, 2024 a rare meteorological condition came across the southern eclipse path with a subtropical jet stream bringing unseasonal middle and high altitude clouds across northern Mexico and southern Texas. At Ciudad Canatlán, a gust of high wind blew the equipment onto the ground and ended data acquisition before totality even began. At El Salto, the middle elevation clouds passed away from the eclipsed Sun five minutes before totality, leaving only areas of high cirrus clouds over most of the sky. This reduced the quality of the acquired data but did not completely stop imaging. Unfortunately, complete cloud cover in Texas resulted in no useful eclipse images at all. The choice of El Salto turned out to be a fortunate one and the results garnered from a subset of the data collected there are described in Section 4.4 below.

2.2 MEE2024 Equipment

The primary components of the stations were the telescopes and cameras, which were supported on a variety of computer-controlled telescope mounts. The team at El Salto used five Tele Vue NP101is telescopes. One of

these, dubbed Station 1, will be the main focus of this paper. It was equipped with a ZWO ASI6200MM Pro wide-field CMOS (complementary metal-oxide-semiconductor) camera. The ASI6200MM Pro is a full-frame camera with ~ 61 Mpx resolution. The other four telescopes, dubbed Stations 2-5, were equipped with ZWO ASI1600MM Pro CMOS cameras. All five stations were mounted on Celestron Advanced VX equatorial mounts and tripods. In order to minimize chromatic aberration and atmospheric refraction and to block some of the short wavelength light from the solar corona, the telescopes were equipped with red filters with a pass band from 550 nm to 700 nm (Edmund Optics, Part no. 89819). Meanwhile, the station at Ciudad Canatlán used an Explore Scientific ED102 refractor with a Player One ZEUS 455M Pro camera.

The telescopes in Texas included two Tele Vue NP101is, two Tele Vue TV-85, one Askar FRA500, one SharpStar 94ED, and one Askar 65PHQ. These were equipped with ZWO ASI1600MM, ASI2600MM, ASI6200MM, ASI183MM and ASI294MM cameras. While eclipse data was not acquired in Texas, calibration images taken on the nights leading up to the eclipse provided a nice data set that allowed us to compare various telescope/camera combinations and suggest the best equipment combinations for future attempts to conduct the Modern Eddington Experiment. See Section 5.2 for details.

2.3 Goals of MEE2024

The main goals of the MEE2024 team during 2024’s total eclipse of the Sun were to:

- Test the viability of using modern CMOS cameras to take eclipse images. Using CMOS sensors instead of CCDs would allow us to use higher frame rates, increasing the number of images taken from dozens to thousands;
- Noticeably improve the statistical significance of the collected data set by adding several additional data collection stations. This would help reduce the uncertainty in our determination of L and create a large data set others could use for educational purposes ([Goldoni & Stefanini, 2020](#));
- Use a large number of students, faculty, and amateur astronomers to test new procedures and calibration techniques that others can implement during future eclipses and to test different imaging mosaics;
- Bring down the risk of clouds by taking data from multiple locations;
- Replace the tedious manual procedure used on 2017 data with software that is open source and provides a complete workflow for analysis of the gravitational deflection of stars near the Sun taken during future eclipses;
- Hone our imaging procedures and parameters to obtain images of stars in the “forbidden zone” (i.e., lying in the region within 1 solar radius from the edge of the disk of the Sun);
- Yield a new measurement of the deflection coefficient L using amateur equipment and test Einstein’s prediction with a level of precision rivaling space-based telescopes and large radio telescope arrays.

The rest of this paper discusses which of these goals we were able to achieve and which fell apart on eclipse day.

3. Summary of Data Collected During MEE2024

Since the Leakey site was clouded out and the station at Ciudad Canatlán was disrupted by wind, data was only collected at the site at El Salto. As discussed in Section 2.2, Station 1 used a full-frame ZWO ASI6200MM camera and Stations 2-5 used ZWO ASI1600MM cameras with smaller 4/3" sensors. In the hours leading up to totality, Station 4 lost polar alignment and would not track properly, so it was unable to obtain useful data. A summary of the data gathered from each of the remaining stations is given in [Table 1](#).

Table 1			
Summary of data acquired at El Salto during MEE2024			
	Images	Exposure [ms]	File Size [GB]
Station 1			
Eclipse Field	124	250	14.4
	124	300	14.4
	123	400	14.3
	124	300	14.4
Zenith Calibration Fields	680	1000	78.9
Station 2			
Eclipse Field	1498	75	46.8
	1478	100	46.2
Zenith Calibration Fields	600	1000	18.8
Station 3			
Eclipse Field	380	1000	11.9

	1202	100	37.6
Zenith Calibration Fields	600	1000	18.8
Station 5			
Eclipse Field	1202	100	37.6
	918	125	28.7
Zenith Calibration Fields	640	1000	20.0
Total:			
Eclipse Fields	7173		266.3
Zenith Calibration Fields	2520		136.5

As a result of the use of modern CMOS cameras with short download times, a total of 7173 images were captured of the eclipse field for a total file size of 266.3 GB. In these, nearly 300 stars could be identified. For comparison, the original 1919 Eddington experiment obtained only two plates with seven stars each and a 1922 experiment collected just “over 100 stars on several plates of glass 14 by 17 inches in size and ¼ inch thick” ([Campbell, 1923](#); [Campbell & Trumpler, 1923](#)). The 1952 attempt, collected only a small number of plates with a few dozen stars ([van Biesbroeck, 1953](#)) with similar numbers of plates and stars obtained during the 1973 experiment ([Jones, 1976](#)). Finally, in 2017 Bruns, using a CCD camera, obtained 45 images with 20 stars and an integrated exposure time of 22 sec ([Bruns, 2018](#)). The size of our data set shows that the data yield of fast CMOS cameras is definitely advantageous.

In addition to the huge collection of eclipse images, 2520 zenith calibration images (136.5 GB) were collected the night before the eclipse. As will be discussed in Section 4.4 below, these data can be used to estimate the optical distortion inherent in the imaging system and correct data taken during the eclipse. Because the collection of data to be processed was so large, the creation of the data processing software described in Section 4 was essential. The best data came from Station 1, so we will use it to discuss the software. This data set and the data sets from Stations 2, 3, and 5 are available for download at <https://www.moderneddingtonexperiment.org>.

4. MEE2024 Data Eclipse Data

b b. bbbb Software and Analysis of

In order to process these large data sets, the MEE2024 data analysis software package was written by A. Smith and D. Smith. Python source code and Windows executables for the package are available for download from GitHub ([Smith & Smith, 2024](#)). The goals for the development of the MEE2024 software package were the following:

1. To develop a robust centroid finder that could operate under a variety of conditions: nighttime imaging, imaging near the full Moon,⁴ and imaging during the total solar eclipse.
2. To provide a method to rapidly and automatically stack a large number of FITS files.
3. To provide a blind plate solver that works for a wide range of fields of view.
4. To perform correction of standard astrometric sources of distortion: stellar aberration, parallax, and atmospheric refraction ([Tatum, 2024](#)).
5. To automatically match detected centroids to the Gaia star catalog ([The Gaia Collaboration, 2016, 2018](#)).
6. To estimate the optical distortion of the telescope using nighttime calibration images by performing a least squares fit between the estimated stellar positions and their established positions in the Gaia catalog.
7. To generate eclipse-day plate scale corrections using fields acquired far from the Sun.
8. To compute stellar displacements from eclipse-day images of star fields taken near the Sun correcting for various known sources of systematic error.
9. To perform a least squares fit to determine the deflection coefficient L .

These features replicate the analysis conducted by Bruns ([2018](#)), who used several commercial software packages (MaxIm DL, Astrometrica, NOVAS, etc) and performed a large part of the analysis manually. While these commercial software tools are quite powerful, it is challenging or impossible to alter the underlying code or sometimes to even fully know what they are doing in detail. The software licenses can also be expensive and the software itself can be difficult to learn, making the analysis challenging for non-expert users. Bruns' analysis also becomes prohibitively time intensive if one is to deal with the large data sets acquired during the 2024 eclipse.

During the development of the MEE2024 software, we used Bruns' calibration and eclipse data from the 2017 eclipse to verify that our software was working correctly. Once it was completed, we were able to do a full analysis of Bruns' data in a few hours (as opposed to the months of work that it took him to analyze the data from the 2017 eclipse) and obtained a result that closely matches his original result as discussed below.

4.1 The MEE2024 Software User Interface

The workflow for the analysis of data outlined above is broken down into three “Tabs” in the user interface (UI) of the MEE2024 software. Tab 1, shown in [Figure 3](#), allows the user to configure and execute preliminary centroid finding, image stacking, and the initial plate solving (steps 1-3 above).

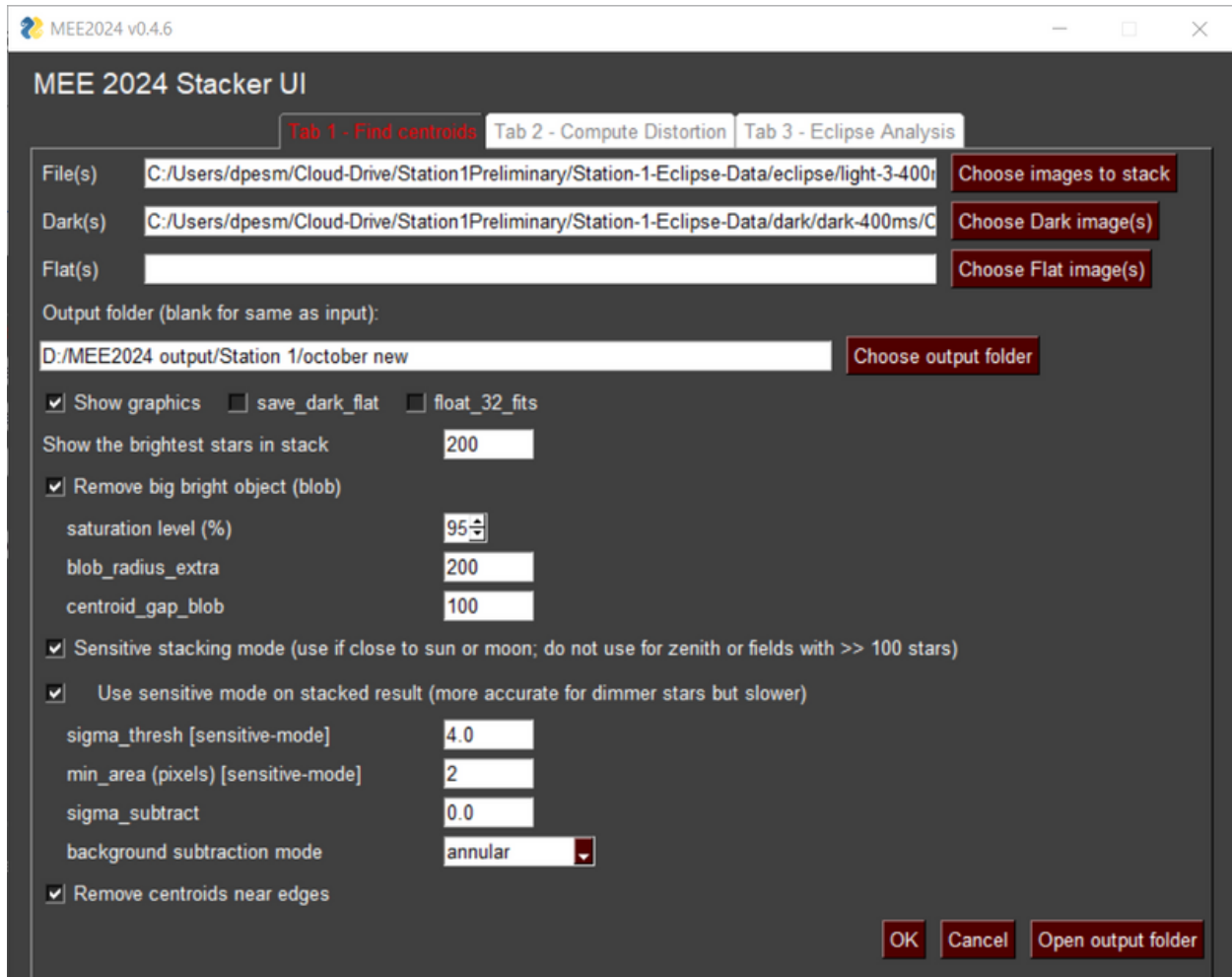


Figure 3

MEE2024 software user interface. Tab 1 is used for finding centroids, stacking, and obtaining a blind plate solution. The code uses standard Python libraries and is open source.

Several sensitivity parameters can be set for the centroid finding. Nighttime data will have very high signal-to-noise since exposure times can be arbitrarily long (e.g. several seconds) and the background light can be quite low. Data taken with a full Moon or eclipsed Sun will have much higher background light so more careful centroid finding is required. In addition, the Moon and Sun need to be appropriately masked or large numbers of spurious centroids will be found. Tab 1 provides several tools to filter out different types of low quality centroids, so that a precise plate solution can be found. Detailed discussions of the various parameters in Tab 1 are provided in the program's GitHub repository ([Smith & Smith, 2024](#)).

Tab 2 of the UI, shown in [Figure 4](#), calculates the standard astronomical distortions ([Tatum, 2024](#)) using Astropy ([The Astropy Collaboration, 2013, 2018, 2022](#)) and uses them to correct the estimated positions of the centroids detected in Tab 1. It then calculates the errors in the corrected centroid positions relative to their positions in the Gaia catalog. Tab 2, then returns the star catalog numbers and their corresponding measured centroid positions, catalog centroid positions, and the discrepancy between the two.⁵ It also returns calibration

coefficients (i.e., higher order plate constants) derived from the star field (steps 4-8 above). Note that since Gaia is able to measure stellar positions to within a few microarcseconds ([Lindgren, 2018](#)), the catalog positions for the stars can essentially be considered the ground truth against which to compare our measurements. Tab 2 offers the possibility to ignore close double-stars and stars that lack a proper motion corrections in the Gaia catalog in the analysis since these are expected to have larger errors, although the number of such stars in a given star field is usually quite small.

MEE 2024 Stacker UI

Tab 1 - Find centroids | **Tab 2 - Compute Distortion** | Tab 3 - Eclipse Analysis

File(s) Choose data (data zip)

Fix distort Choose distortion files

Fix order higher than

Output folder (blank for same as input): Choose output folder

☒ Show graphics

Maximum star magnitude

☐ Guess Date!

Observation Date UTC (YYYY-MM-DD)

Distortion fit tolerance (arcsec)

Distortion polynomial order

Rough fit threshold (arcsec)

☐ Crop circle? (input 0.0 to 1.0)

☒ Discard double-stars or stars with missing proper-motion

Corrections for aberration, parallax, and refraction:

☒ Enable aberration and parallax?

☐ Enable gravitational correction?

☒ Simultaneous deflection constant and platescale fit

Observation Time UTC (hh:mm:ss)

Observation Latitude (degrees)

Observation Longitude (degrees)

☒ Enable refraction correction?

(optional) Observation temp. (°C)

(optional) Observation pressure (millibars)

(optional) Observation humidity (0.0 to 1.0)

(optional) Observation height (m)

(optional) Observation wavelength (μm)

OK Cancel Open output folder

Figure 4

MEE2024 software user interface. Tab 2 is used to perform standard astrometric corrections using AstroPy and characterizes the optical distortions of the telescope system. It can also be used to apply the corrections derived from calibration frames to the eclipse frames. The output of Tab 1 is the input to Tab 2.

The coefficients computed using Tab 2 allow users to correct centroids from images gathered during the eclipse for optical distortions in the imaging system. Following Bruns (2018), calibration coefficients are first calculated for images taken the night before the eclipse or far away from the Sun on eclipse day. The output

from Tab 2 from the calibration observations can then be used as an input to fix the higher order plate constants for the analysis of images taken near the disk of the Sun during the eclipse. As previously described by Bruns (2018), most good quality telescopes can be corrected sufficiently with a cubic polynomial function. However, as we discuss in Section 4.4, telescopes with very wide fields of view can have a significant quintic distortion, which also needs to be accounted for.

Finally, Tab 3 (see Figure 5) uses the errors in stellar positions computed in Tab 2 to compute L by using a least squares method (step 9 above). Three “data limits” are included in Tab 3. The radial cutoff is meant for extremely wide fields of view where the edges of the frame can have significant errors. The stellar magnitude limit is effectively a limit on the acceptable signal-to-noise of the centroids since higher magnitude (dimmer) stars are more difficult to locate precisely. Finally, the user has the option to exclude double stars, which are unlikely to affect the analysis of calibration fields where there are a lot of stars but can potentially impact the analysis of eclipse fields. Tab 3 also provides an option to perform the analysis for images taken during a full Moon. Section 4.2, below, provides a more detailed walk-through of how the software works using Bruns’ (2018) data from the 2017 eclipse.

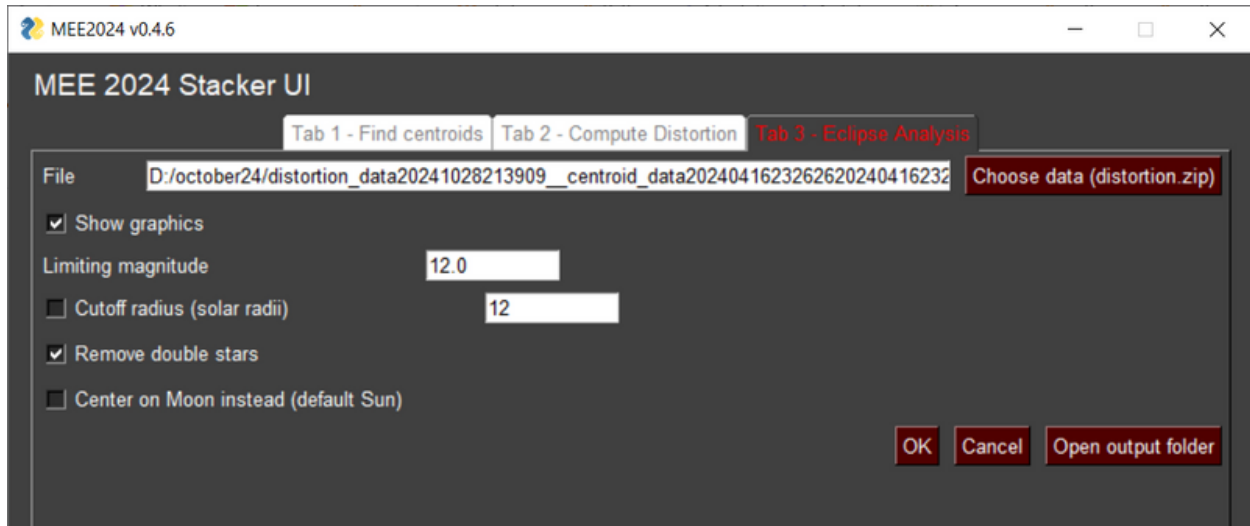


Figure 5

MEE2024 software user interface. Tab 3 primarily produces graphical output (see, e.g., Figure 8) from the numerical output of Tab 2. This section of the software calculates the deflection coefficient L by fitting the star deflections to a $1/R$ curve using a least squares method.

4.2 Detailed Workflow for the MEE2024 software

When analyzing eclipse data, the first step is to generate the nighttime distortion files. These are sometimes referred to as “zenith calibrations”, although they need not be taken exactly at the zenith. A series of around a dozen fields taken near the zenith (to minimize atmospheric refraction/distortion) are first stacked and plate solved in Tab 1. These results are then passed through Tab 2 and, after being corrected for standard astrometric

distortions, errors in the observed positions of stars relative to their catalog positions are generated. These error fields are then fit to polynomial fitting functions to compute higher order plate constants (see Equation 2 below). The resulting distortion calibration files can then be used to correct the measured positions of stars in other images for optical distortions in the imaging system (see [Figure 10](#) and Section 4.4 for details). The software allows for corrections of optical distortion of up to septic (i.e., seventh) order. Nighttime images, where lots of high quality centroids can be detected, are ideal for determining third and higher order correction coefficients. These coefficients can then be used to make these higher order corrections to full Moon or eclipse-day images where high quality centroids are more sparse.

In Bruns’ 2017 eclipse data ([2018](#)), right and left calibration fields were observed on the day of the eclipse. The resulting pair of stacked images was then corrected using the set of zenith calibration files. In this second pass through Tab 2, “Fix order higher than” is set to “quadratic”. This means that the cubic and higher coefficients are frozen at the nighttime readings, but the linear and quadratic “plate scale distortions” (typically due to thermal expansion) are recalculated. A pair of new calibration files was generated and these were used on the stacked eclipse field data. To do this, Tab 2 was run on the eclipse day but “Fix order higher than” was now set to “constant”. In this case, “constant” corresponds to a translation in x - y pixel space.

Note that the number of eclipse-day calibration fields could be one rather than two or, in theory, it could be more than two. Generally, an even number of calibration fields taken symmetrically before and after the eclipse is expected to give the best correction. If no eclipse-day calibration data is available, one cannot simply use the linear and quadratic plate scale from the nighttime calibrations. This special case needs to be dealt with in a different way that will be discussed in Section 4.4 below. The final output data file of Tab 2 is then entered into Tab 3 (see [Figure 5](#)), which performs the gravitational distortion analysis.

In Tab 2 there is a “Maximum star magnitude” feature that is used for the calibration process. For nighttime calibration, a limiting magnitude of around 12 is a reasonable choice: it gives a good number of stars to generate a distortion calibration field. Adding more, but dimmer, stars could result in a worse final rms error. The same logic appears to be true for the eclipse field. Including dimmer stars means there are more stars to fit, thereby decreasing the statistical error. However, if one were to weight all the stars equally then including many low signal-to-noise centroids from dim stars means the high signal-to-noise data points are “diluted”. For both calibration and eclipse fields, there is typically a “limiting magnitude” which minimizes total residual rms error. Ideally, non-equal weights could be assigned to stars based on an estimate of their signal-to-noise ratio, but an unbiased implementation of this method is non-trivial.

Tab 3 implements the method used by Bruns ([2018](#)) to analyze the 2017 eclipse data and performs a least squares fit of the stellar deflections to a $1/R$ function using the deflection coefficient L and the position of the center of the eclipse image field (right ascension RA, declination DEC, and roll) as parameters. The reason for fitting the image center position is that the plate solver is not able to exactly find the center position; i.e. there is a very small initial offset error in the displacement measurements with respect to the center of the Sun.

4.3 Reanalysis of Bruns' Data from 2017

In order to validate the software, we used it to analyze Bruns' data from 2017 and found close agreement with his original results. [Figure 6](#) is a screenshot of the initial representation of the eclipse deflections. In the program, it is actually possible to zoom in on individual stars and see that there is an apparent outward shift of the stars compared to their catalog positions (see [Figure 1](#)). In this example, we have used a limiting magnitude of 10.0, which results in 17 stars in the field of view (Brun's used 20 stars). The Gaia catalog number and the stellar magnitude are noted next to each centroid. The two stars closest to the Sun (with magnitudes of 7.1 and 7.5) are among the three brightest stars in the whole field.

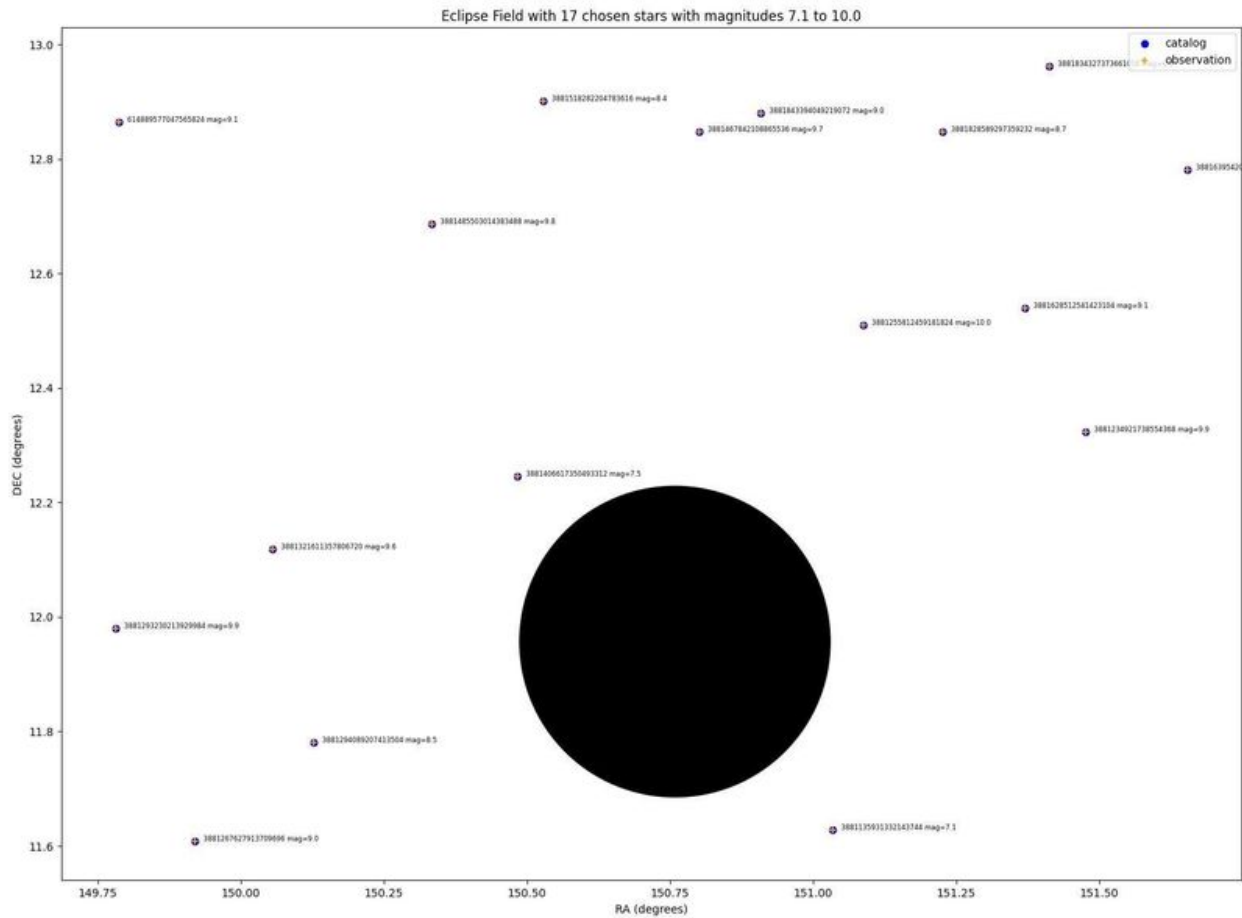


Figure 6

MEE2024 software graphical output of star locations from the 2017 Bruns data. The experimental data is compared to the Gaia star catalog. Here, the vertical axis is the declination (DEC) of the stars and the horizontal axis is their right ascension (RA). The large black circle is the disk of the Sun. The labels next to each star gives their Gaia catalog number and stellar magnitude.

[Figure 7](#) shows how the program uses a least squares analysis to find that the minimum aggregate error occurs at $L = 1.74425$ arcsec, only 0.4% from Einstein's prediction. A "naive uncertainty estimate" can be defined as

the residual rms error in the centroid positions divided by the square root of the number of stars. In this case, that error is 2.1%, similar to the uncertainty calculated by Bruns (2018), giving us a best estimate of $L = 1.74 \pm 0.04$ arcsec. This is consistent with Bruns' result of 1.75 ± 0.05 arcsec and with Einstein's prediction.

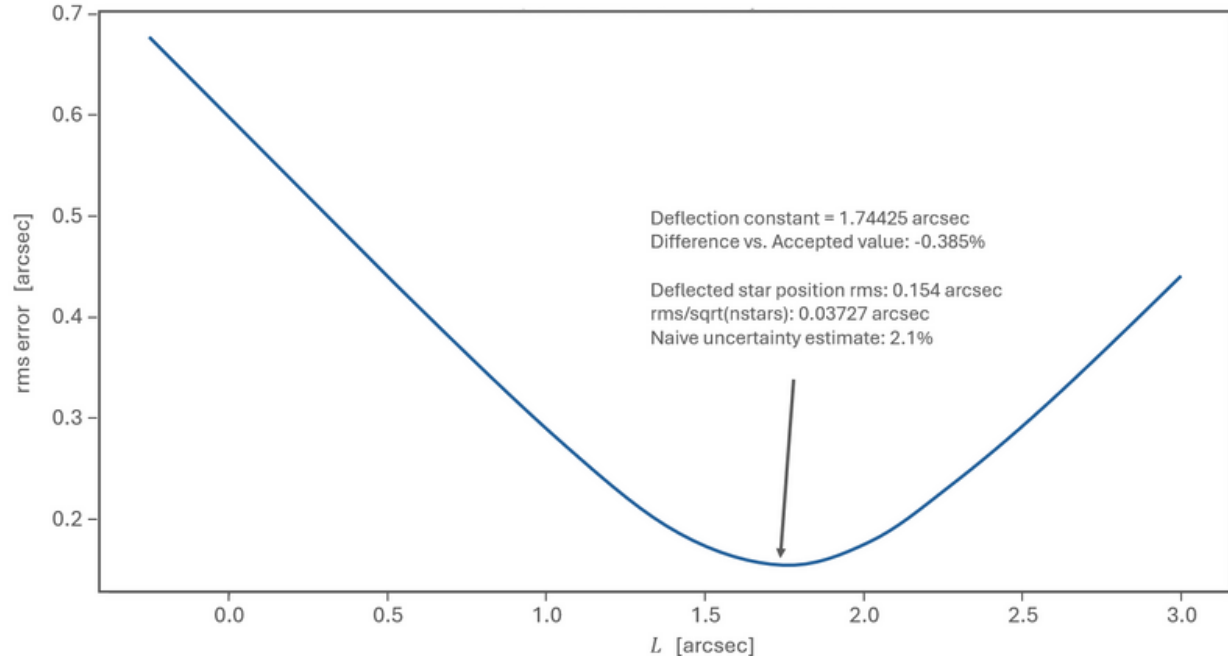


Figure 7

MEE2024 software graphical output that shows how Einstein's deflection coefficient L is found by minimizing the error in fitting Bruns' 2017 data to a hyperbolic $1/R$ function. The vertical axis is the rms residual error in arcseconds and the horizontal axis is the deflection coefficient L . Using our software to analyze Bruns' data set from 2017, we calculated that

$$L = 1.74 \pm 0.04 \text{ arcsec.}$$

Finally, Tab 3 of the program generates a plot of the radial deflection of the stars versus their radial distance from the Sun's center and overlays the best fit $1/R$ curve. An example of the program's output is shown in [Figure 8](#).

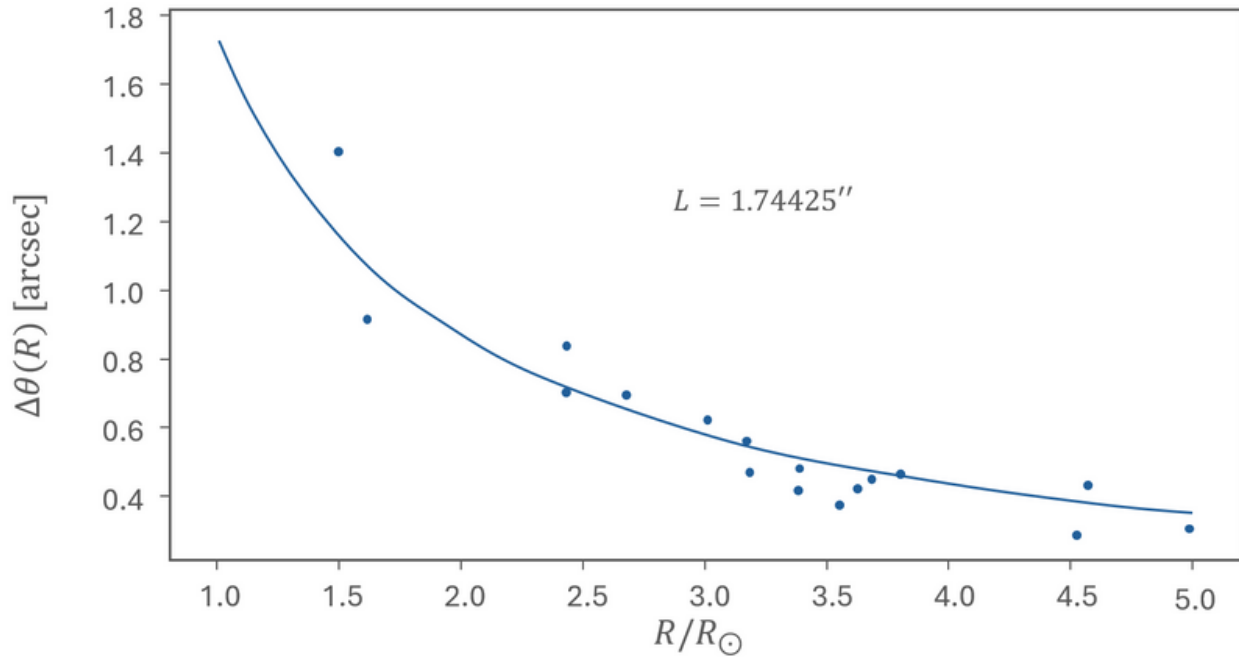


Figure 8

Sample MEE2024 software graphical output plotting the radial deflections $\Delta\theta$ as a function of radial position in solar radii R , overlaid with the curve of best fit to Equation 1 for Bruns' data from 2017.

4.4 Analysis of MEE2024 Data

We will now discuss our analysis of data taken during the 2024 eclipse. The analysis presented here is limited to a subset of the Station 1 data obtained by team members Delgado, Matin, and Dittrich: the 123 eclipse images taken at 400 ms exposure on April 8th, 2024 and 17 zenith fields captured the night before. Each zenith field is a stack of twenty 3 sec exposures. The total integrated eclipse exposure is 49.2 sec. [Figure 9](#) gives an example of one zenith stack analyzed using Tab 1 of the MEE2024 software.

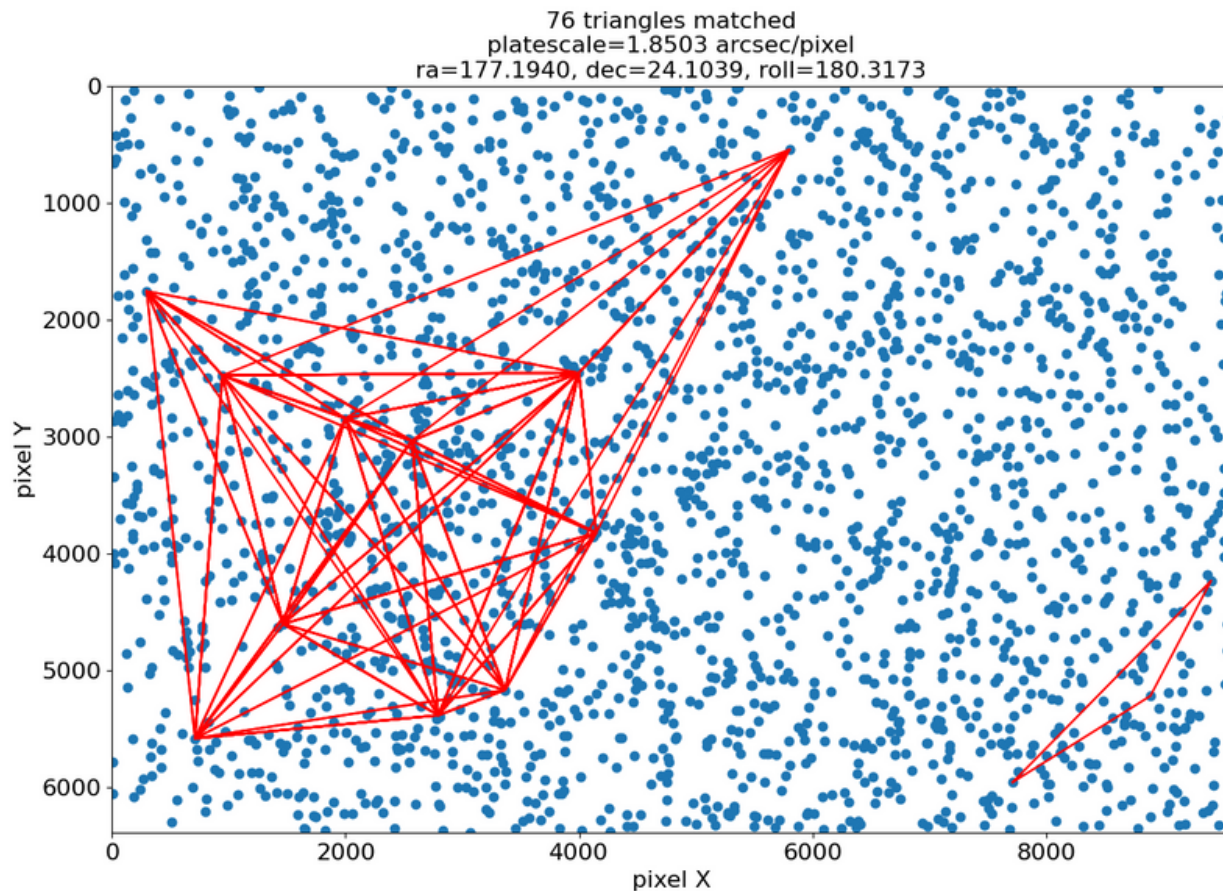


Figure 9

Sample graphical output from the MEE2024 software package's Tab 1 for a zenith calibration set from Station 1 at the El Salto site. Triangles (shown in red) formed from detected centroids (shown as blue dots) are matched against triangles computed from the Tycho-2 star catalog to determine the field coordinates and plate scale. The blind plate solver was specially developed for the MEE2024 software package and is very quick and quite robust. The computed plate scale, right ascension, declination, and roll are displayed at the top of the image. The plate solver uses a built-in database so it does not need an internet connection, which is useful when testing acquisition systems in the field.

In this calibration field, 2383 centroids were found. The blind plate solver matched 76 triangles in order to find the image field coordinates and plate scale. The centroid data was then entered into Tab 2 and a full quintic distortion correction was computed. In this case, a limiting magnitude of 15 was used in order to get a very high density of stars in the field of view. The more centroids that can be detected in the zenith field, the more accurate the estimate of the optical distortion of the telescope system will be. A high density of centroids is particularly important if the telescope has a large and complex distortion as might be expected for Station 1 since it used a full-frame camera and a large field of view of about 4.9° by 3.3° .

To appreciate how much quintic distortion affects Station 1, [Figure 10](#) shows a graphical representation of the correction function computed for the zenith data shown in [Figure 9](#) with the residuals for the computed

centroid positions overlaid. The fit is extremely good, except perhaps in the extreme corners, where the distortion is particularly high.

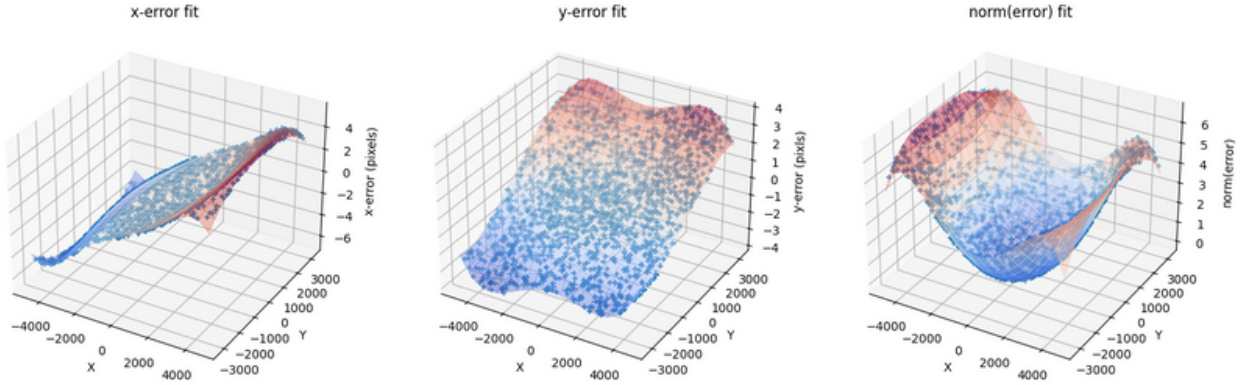


Figure 10

Sample graphical output from the MEE2024 software package's Tab 2. These plots show the correction function computed for Station 1. (Left) The blue dots indicate the measured residuals Δx in the x -component of the experimentally determined stellar positions when compared to the Gaia catalog as a function of their position in pixels relative to the center of the image. The surface shows the surface of best fit, which can be used to compensate for the distortion when measuring star positions in later experiments. (Middle) The residuals in the y -position of stars Δy along with the surface of best fit. (Right) The norm of the overall residuals $\Delta s = \sqrt{\Delta x^2 + \Delta y^2}$ as a function of position. It can be seen that the correction function can correct stellar positions well throughout the full image. While most systems can generally be corrected with a cubic function, the wide-field of Station 1 requires a quintic function as can be seen near the corners of the image.

[Figure 11](#) provides a representation of the polynomial coefficients which correspond to the distortion surface plotted in [Figure 10](#) and how these vary over many zenith calibration sets. The form of the corrections are the two polynomials

$$\begin{aligned} x_c &= x + \sum_{n=0}^N \sum_{i=0}^n A_{n,i} \alpha^i \beta^{n-i}, \\ y_c &= y + \sum_{n=0}^N \sum_{i=0}^n B_{n,i} \alpha^i \beta^{n-i}, \end{aligned} \tag{2}$$

where (x, y) are the initial observed centroid coordinates, (x_c, y_c) are the corrected pixel positions, while (α, β) are the shifted and scaled coordinates such that the center of the image is $(0, 0)$ and the long edge is at ± 1 . An advantage of using scaled coordinates is that the magnitudes of the coefficients are of order unity and can be compared easily.

The coefficients $A_{n,i}$ and $B_{n,i}$ weight terms up to finite order N , typically either cubic ($N = 3$) or quintic ($N = 5$). If one assumes there to be a perfect mirror symmetry in the distortion along the x and y axes, then only the odd order coefficients would be non-zero ($n = 1, 3, 5, \dots$) and for the $A_{n,i}$ only odd i are allowed, while for the $B_{n,i}$ only even i are allowed. The translational, rotational, and scaling degrees of freedom (i.e. RA, DEC, roll, and plate scale) allow for four degrees of freedom to be removed from the polynomials; we choose the convention of having $A_{0,0} = B_{0,0} = A_{1,1} = 0$ and $A_{1,0} = -B_{1,1}$, which means that the distortion coefficients can be uniquely determined. We observe that the mirror symmetry rule is quite well-respected empirically, except for the linear and quadratic orders, where small asymmetries in the instrument (such as lens alignment) result in non-negligible terms. We suggest it may be acceptable to set all “symmetry forbidden” coefficients (except for the linear and quadratic ones) to zero, which would eliminate about $\frac{3}{4}$ of the degrees of freedom in the distortion fit.

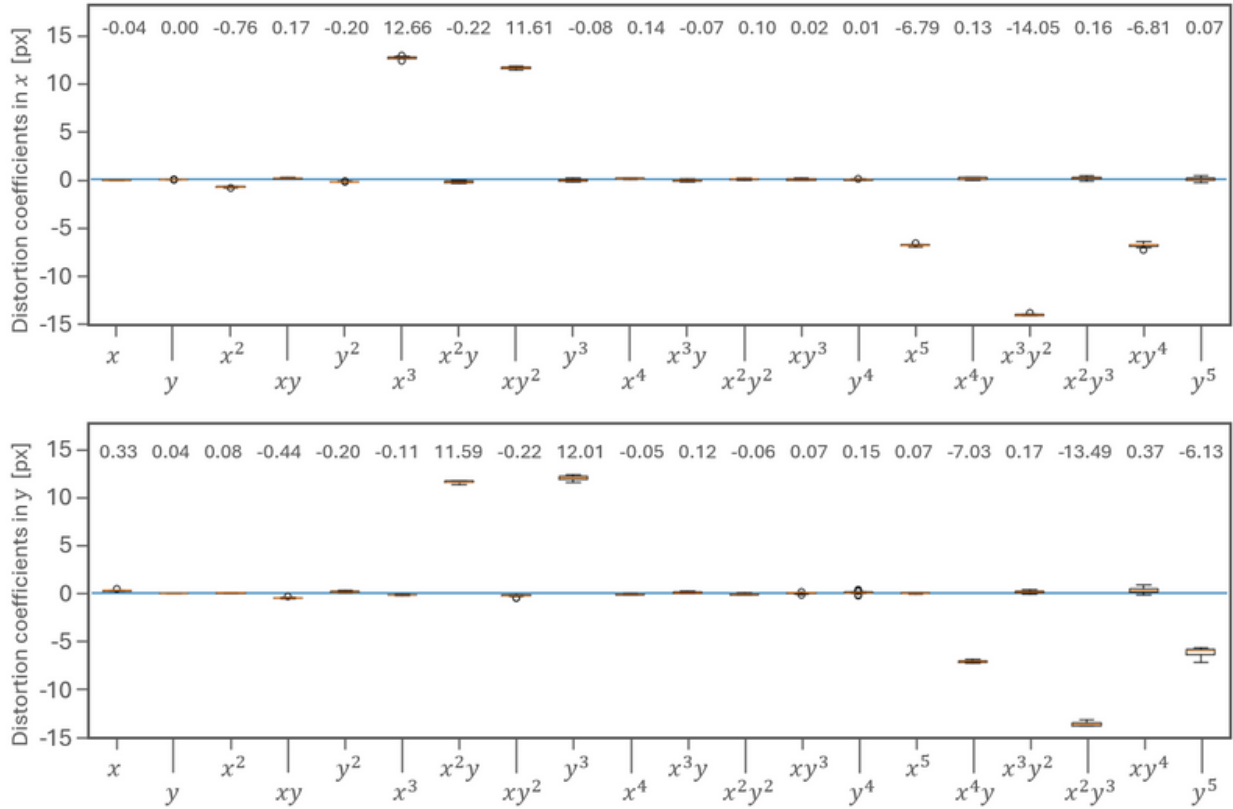


Figure 11

Graphical representation of the distortion coefficients for Station 1. The top panel shows the coefficients for the x -correction ($A_{n,i}$), while the bottom panel corresponds to the corrections in y ($B_{n,i}$). Note that the cubic and quintic components are dominant. The quartic components are negligible, as expected by symmetry. The quadratic components are small but meaningful. The spread of values measured for 17 separate zenith calibrations is shown, demonstrating excellent stability in the fit. The mean value of each coefficient averaged over the 17 data sets is printed along the top.

[Figure 12](#) provides a few informative plots of the residuals obtained by comparing the experimentally measured positions of stars from zenith calibration images to their known positions from the Gaia catalog ([The Gaia Collaboration, 2016, 2018](#)). As can be seen, the residuals are fairly insensitive to a variety of factors, including the star magnitude, parallax, and radial distance from the center of the image.

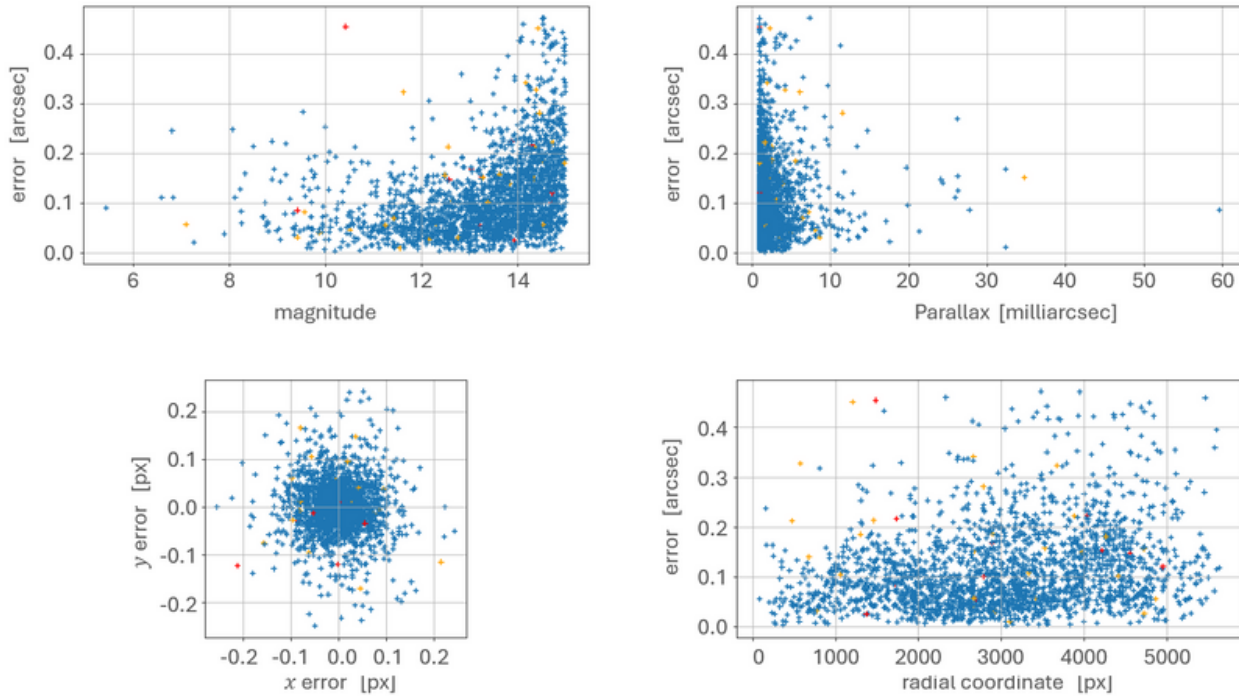


Figure 12

Sample graphical output from the MEE2024 software package’s Tab 2 showing scatter plots to visualize the random (and potentially systematic) errors in a zenith calibration from Station 1. Stars for which proper motion information is missing in the Gaia catalog are highlighted in red. Double stars, which can potentially lead to centroids with higher error, are marked in orange. (Top left) Residual error as a function of star magnitude. The residual error increases significantly for stars with magnitudes greater than ~ 13 ; there are also many more dim stars (mag 14) than bright ones (mag 7). (Top right) Residual error as a function of parallax angle. There is very little dependence of the residuals on parallax angle (i.e., stellar distance from the Sun) (Bottom left) The scatter plot of the stellar residuals in x and y is symmetric (helping to exclude the possibility of a systematic bias) and generally shows that the rms positions of the centroids can be corrected to within 0.14 pixels of their theoretical catalog positions. (Bottom right) A plot of the residuals as a function of radial distance from the center of the image shows that the calibration works well all the way to the edge of the image.

Having computed higher order plate constants from the zenith images, we proceeded to analyze the eclipse-day data. The stacking of the 123 frames taken at 400 ms exposure was done similarly to the validation run done on Bruns’ 2017 dataset discussed in Section 4.2. [Figure 13](#) shows the stars used for stacking and the number of times a star appeared in the 123-frame dataset (122 is the maximum times a match can occur with respect to a reference frame):

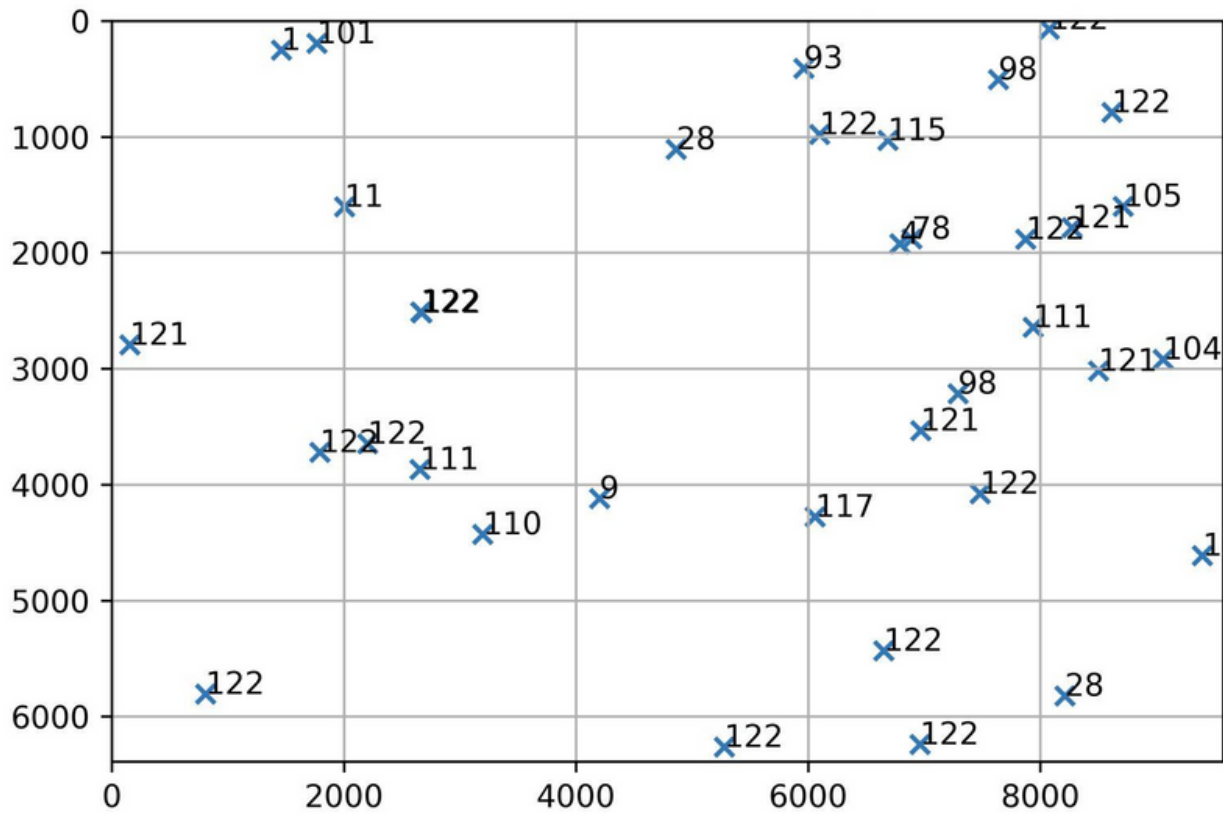


Figure 13

Sample graphical output from Tab 1 of the MEE2024 software package's multi-star stacking algorithm. Each cross marks the position of a centroid in x - y pixel space. The number next to each centroid gives the number of images in which the centroid was found. These centroids are then used to stack the images using a multi-star method. In our experience this gives better results and is more robust than a single-star method.

The program then proceeded to plate solve the stacked image and was able to correctly identify the eclipse location. 276 centroids were found in the stacked image and the program's built-in star catalog (derived from the Tycho-2 survey, see [Høg et al., 2000](#)) could identify 99 with brightness down to magnitude 11.

One of the objectives of MEE2024 was to test the feasibility of using a full-frame camera sensor to capture images of stars over a wide field of view. This avoids slewing the telescope between different fields, which can add complexity and consume valuable observation time. In Bruns' 2017 measurements, the telescope slewed to either side of the eclipse field where gravitational deflections in stellar positions were small and these calibration fields are used to correct the linear and quadratic distortion terms of the telescope system under similar thermal and orientational conditions to when the eclipse field is imaged.

Since in MEE2024 Station 1 did not slew during the eclipse, no eclipse-day calibration data were available to refine the linear and quadratic distortion terms from their nighttime values. To get around this, we chose to use

the quadratic, cubic, quartic and quintic terms of the zenith calibrations taken the night before, leaving only the linear terms to be adjusted from the eclipse star field. The rationale behind this is that the linear plate scale terms are most sensitive to the thermal difference between the nighttime calibration and the daytime measurement.

As pointed out by Freundlich and Lederman (1944), it is highly desirable to have an independent determination of the plate scale: “The inclusion of a scale correction S and of L [the plate scale and the deflection coefficient, respectively] as unknowns in the Least Square solution has to be avoided. . . The scale correction has therefore to be determined separately.” In the case where the plate scale and the deflection coefficient are both unknowns in the least squares solution, then the largest possible number of stars should be used. Fortunately, the stack of 400 ms exposures from Station 1 contained a relatively large number of stars.

Mikhailov (1959) discussed the eclipse data taken by various groups between 1919 and 1952. It was common in these early experiments to have no independent plate scale since the telescopes were too large to be slewed. Mikhailov describes an additional shift of the stars in the radial direction that can be introduced by the plate scale uncertainty. Therefore the expression for the radial displacement of each star become $\Delta\theta = L/R + SR$, where the two constants L and S are to be determined by a least squares method. The constant S is mainly obtained from stars with large values of R , while L is determined mainly from stars with small values of R . The distant stars should be at least 6 to 8 solar radii away, requiring a field of view of at least 4 degrees.

In a sense, MEE2024 Station 1 was a hybrid experiment: modern in its acquisition technology but classical in its data taking strategy. The analysis was therefore also hybrid: done along the lines presented by Mikhailov (1959) but using comparisons to the Gaia star catalog. As shown in Figure 14, Station 1 could observe a field going out to 11 solar radii with over 170 stars. This field is as large as the historical experiments but with many more visible stars (see Table I in Mikhailov (1959)). As a result, despite not having an external plate scale calibration, we believe that we were able to determine L to within 5% of its accepted value. Nevertheless, we believe that even higher accuracy could have been achieved if at least one external calibration field had been taken.

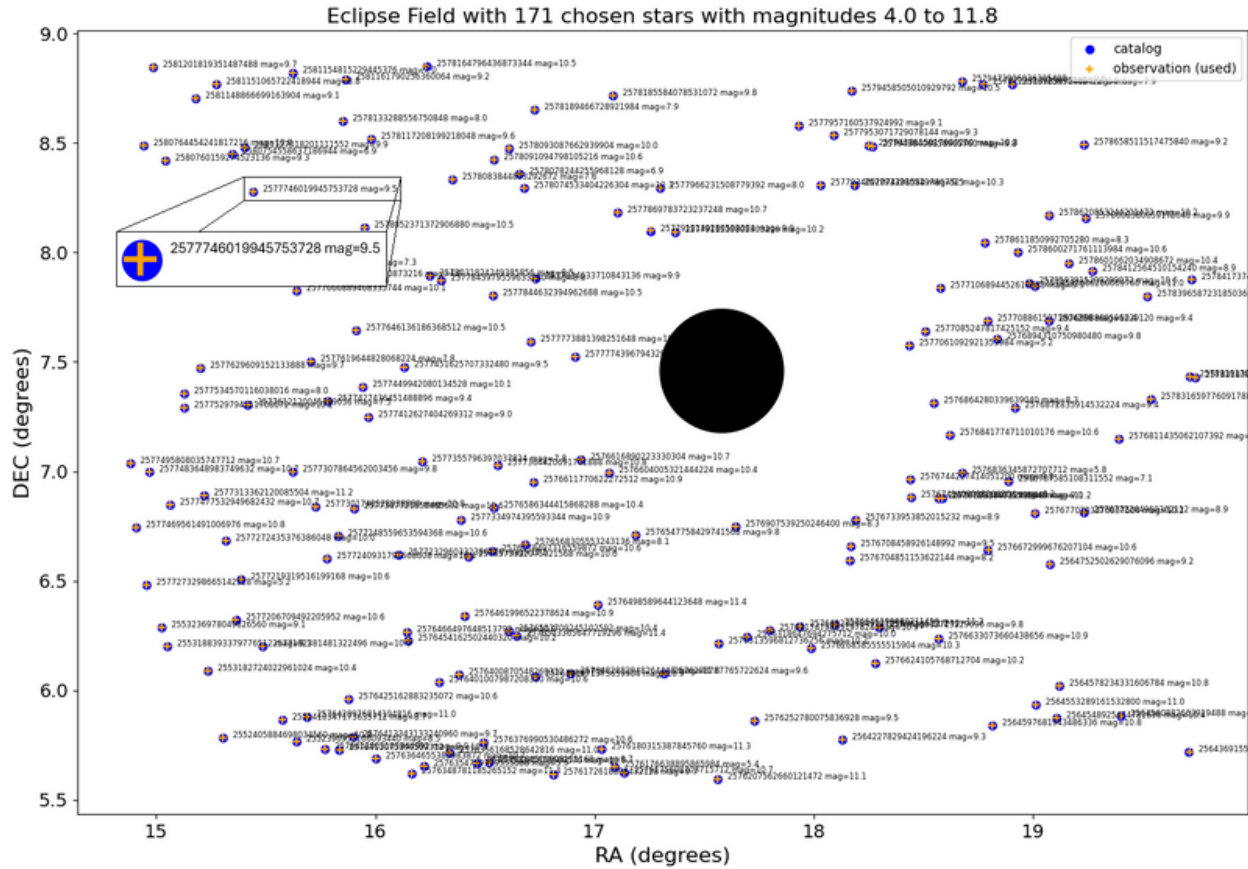


Figure 14

Sample graphical output from Tab 2 of the MEE2024 software package showing the star deflections measured during MEE2024 from the Station 1 data. These deflections were computed from a data set of 123 frames taken at 400 ms exposure for a total exposure of 49.2 sec. Stellar with magnitudes down to 12 were used. The inset shows that, after performing standard astrometric corrections and correcting for distortion, the observed position (orange cross) of Gaia 2577746019945753728 is pushed radially outward from its catalog position (blue circle). The large black circle marks the disk of the Sun.

To get a large number of stars, we used a limiting magnitude of 12 for the eclipse-day data. As mentioned above, magnitude 12 is a good value for nighttime calibrations and we would normally suggest limiting the eclipse-day stars to between magnitude 10 and 11. As shown in the top left part of [Figure 12](#), the error in star location starts to rise considerably above magnitude 13, so we excluded these stars from our analysis. We also used the double star removal option since these stars are expected to have particularly large errors.

In order to not exclude any more centroids, we used a large “Distortion fit tolerance” of 2.0 arcsec and a large “Rough fit threshold” of 100 arcsec (see [Figure 4](#)) in Tab 2. We also avoided using the “Crop circle” option, which would have excluded stars in the corners. We think the quintic distortion correction was sufficient at the corners as the error at the edge of the frame was not noticeably worse than closer to the middle (bottom right

panel of [Figure 12](#)). When computing L in Tab 3, we also avoided limiting the magnitude of the stars or using a cutoff radius (see [Figure 5](#)).

Once the centroids were matched to the Gaia star catalog and the positional errors found, a statistical method was used to vary the plate scale and the deflection coefficient. Importantly, the toggle for “Simultaneous deflection constant and plate scale fit” in Tab 2 was selected (see [Figure 4](#)). After an approximate plate scale was found from Tab 2, Tab 3 was used to perform a least squares regression on the observed radial deflections from the Sun using Mikhailov’s equation $\Delta R = A/R + BR$. This effectively allowed a simultaneous refinement of the plate scale measurement and the deflection coefficient. Using this method, the standard statistical Bayesian uncertainty and covariance of the plate scale and deflection coefficient could be easily computed. We found that the estimated errors in L and plate scale had a strong negative correlation (correlation coefficient = -0.783), as can be seen in [Figure 15](#). This intuitively represents the fact that a decrease in plate scale results in an apparent displacement similar in effect to an increase in deflection coefficient. If the plate scale was known exactly, the uncertainty in L could be reduced by around a factor of two, demonstrating the importance of taking calibration fields.

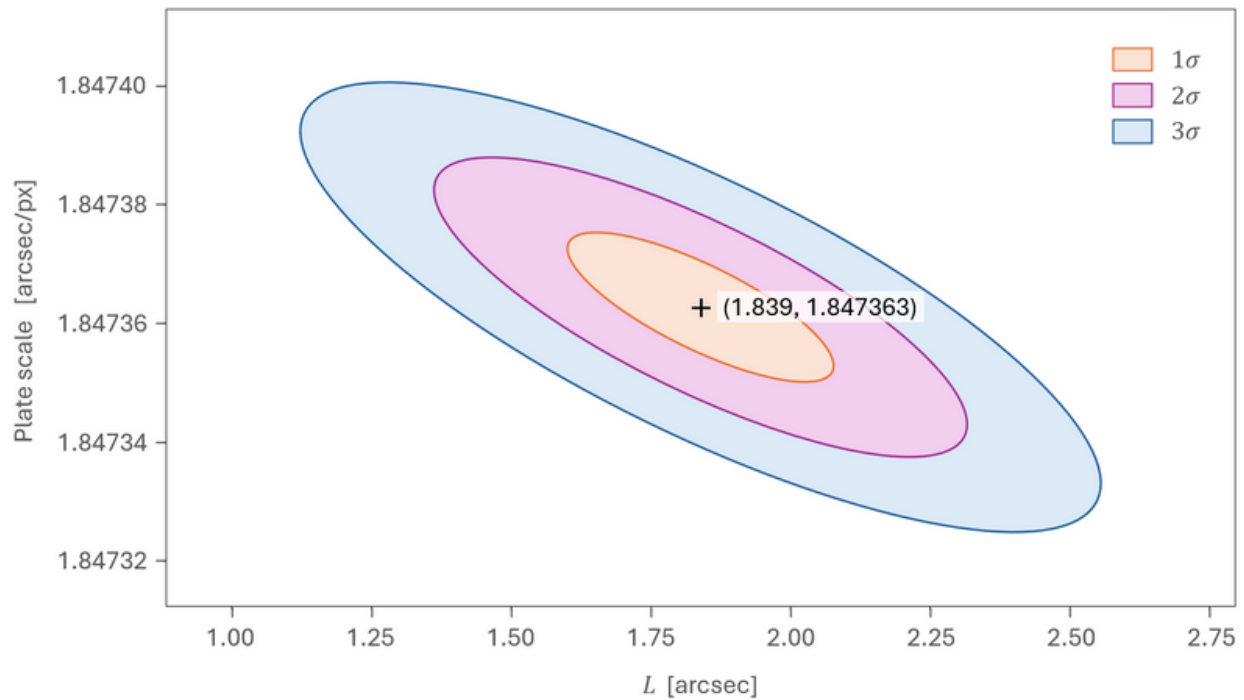


Figure 15

Covariance method for determining the deflection coefficient with an uncertain plate scale. For the Station 1 data, a value of $L = 1.839$ arcsec was found, with a 1 sigma error of 0.239 arcsec. The plate scale result is 1.847363 arcsec/pixel, with a 1 sigma error of 1.3×10^{-5} arcsec/pixel. The deflection coefficient and the plate scale are negatively correlated (correlation coefficient = -0.783) for reasons discussed in the text.

The value of the deflection coefficient of $L = 1.839$ arcsec is about 5% greater than the accepted value of 1.7512 arcsec. Interestingly, when Bruns (2018) analyzed his 2017 data using only the eclipse field (ignoring the independent plate scale derived from the two calibration fields), he found $L = 1.86$ arcsec, an error of 6%. From Table I of Mikhailov (1959), the most similar historical experiment to Station 1 is the 1922 Lick Observatory expedition to Australia. Mikhailov’s reanalysis of the Lick data showed 71 stars at distances of 2.1 to 13.0 solar radii, with $L = 1.83$ arcsec. The clustering of these three calculations within a range of 0.03 arcsec is intriguing and may suggest a systematic error in this method of data analysis.

When we use the value of L found in Figure 15 and re-plot the radial deflections versus distance from the Sun, we get the results found in Figure 16. It is immediately apparent that the scatter in the data points is quite large, especially when compared to Figure 8, a similar plot of Bruns’ 2017 data. Observers at the site in Mexico noted that there was high wind and intermittent clouds, which may have caused a much higher level of air turbulence and therefore higher centroid error than was seen in the nighttime zenith calibrations when the air was very still. Nevertheless, the very large dataset of 171 stars appears to allow averaging out the large random errors to arrive at a value of L that is off by only a few percent.

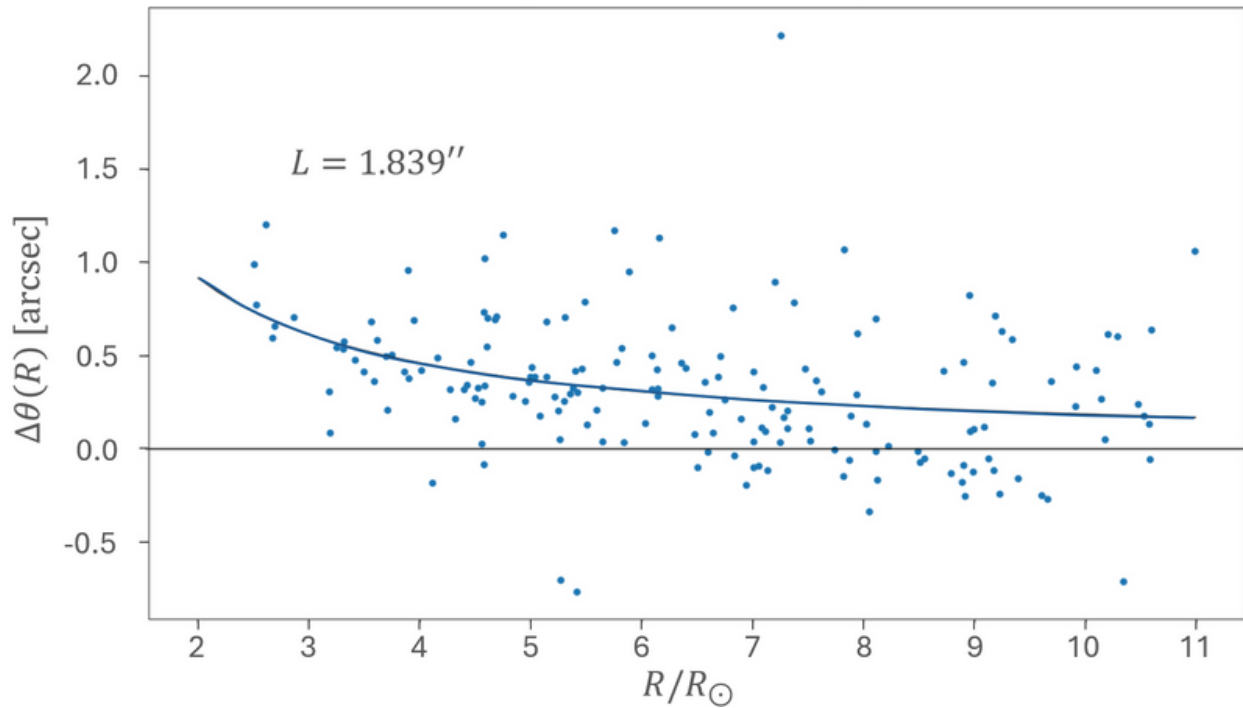


Figure 16

The plot of radial deflections $\Delta\theta$ versus radial position R for 171 stars with magnitudes between 4.0 and 11.8 shows a large amount of scatter and no stars closer than 2.5 solar radii for data collected during the eclipse of April 8th, 2024. For a similar plot using Bruns’ data from 2017, see Figure 8.

Note also that there were no stars detected at less than 2.5 solar radii, so stars with very large deflections were not in the 400 ms dataset. We did an analysis of the 250 ms Station 1 dataset to see if any closer stars could be

identified. One close star, Gaia 2577063532462714368 with magnitude 9.7, was found. This relatively dim star at a distance of $1.97R_{\odot}$ from the disk of the Sun was just visible near the solar corona at this shorter exposure. However, the signal-to-noise was quite poor and the deflection error was around 1 arcsec more than would be expected. The addition of this single star close to the Sun throws off the whole calculation of L since inner stars have much higher influence on the calculation. We therefore conclude that stars close to the corona also need to be quite bright in order to be accurately located. A single close-in outlier can actually cause a significant increase in error. In the case of Bruns' 2017 data, the two inner stars had magnitudes of 7.1 and 7.5 and were about a factor of 10 brighter than Gaia 2577063532462714368.

5. Recommendations for Future Experiments

As we have shown, it is possible to measure Einstein's deflection coefficient using commercially available refractors and CMOS cameras if careful procedures are followed. Following Bruns' (2018) approach, calibration images taken at night were used to determine the higher order optical distortion coefficients of the telescope and camera system. Those coefficients were then applied to correct eclipse-day images and measure stellar positions with sub-arcsecond precision. As we will show below, these optical distortion coefficients are stable over several weeks. There are also additional lessons for researchers attempting to repeat the Modern Eddington Experiment that can be learned by analyzing the regions of the images that have the lowest level of distortion and by analyzing the star fields that will be near the Sun's disk during future eclipses.

5.1 Stability of Distortion Measurements

The cubic optical distortions of two refractors were repeatedly measured over several clear nights during the spring of 2024. A Tele Vue TV-85 with an $0.8\times$ reducer was paired with a ZWO ASI1600MM camera and a Tele Vue NP101is was paired with a ZWO ASI2600MM. The resulting cubic distortion computed near the edge of the images for the two systems are plotted in [Figure 17](#). The mean values for the two telescopes over all nights were 7.09 ± 0.06 arcsec and -1.03 ± 0.03 arcsec, respectively. These data show that the cubic optical distortions are stable and can be measured to within a few percent. The ~ 1 -3% stability of the cubic distortion of both telescopes is expected to apply to the other similar telescope designs, so the suitability of a particular telescope/camera combination can be evaluated well before the eclipse.

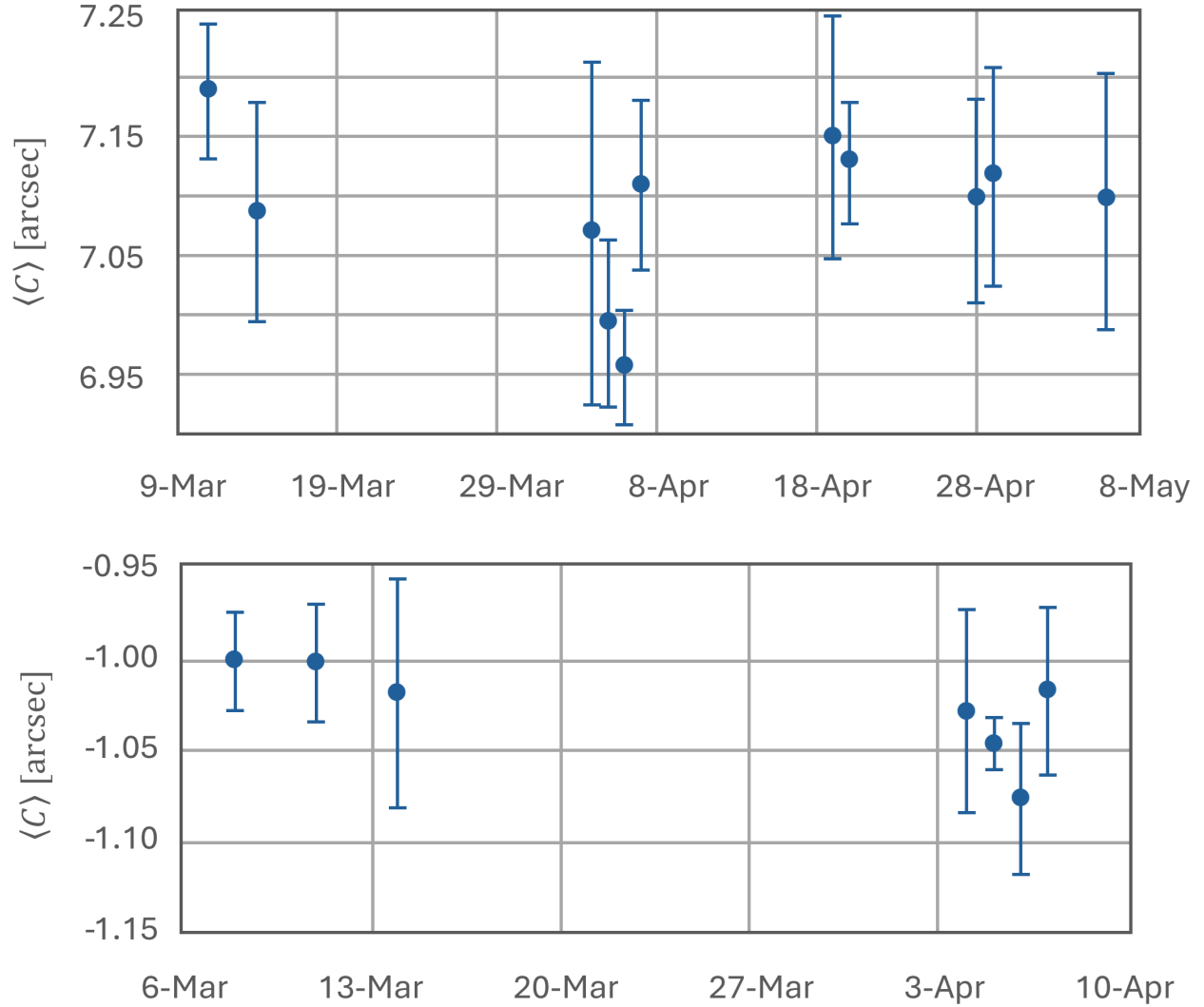


Figure 17

The mean values of cubic optical distortions C evaluated at a point 95% of the way to the edge of the image along the x -axis averaged over several acquisition sessions for (top) the Tele Vue TV-85 telescope with a 0.8 $\times$ reducer and (bottom) the Tele Vue NP101is telescope. The dots indicate the average of C taken over several star fields in a given night and the error bars its standard deviation.

5.2 Optimization of the Imaging Configuration

In the previous MEE expeditions, the Sun was usually placed near the center of the image. Upon further review, this wastes much of the camera sensor. In that configuration, stars near the image center are not detectable, yet this is also the part of the image where the cubic optical distortions are the smallest. On the other hand, stars near the edges of the image have the smallest deflections and the largest distortions.

Furthermore, distortion shifts orthogonal to the deflection shifts can be ignored, which is something to take advantage of if the maximum usable area is to be optimized.

Based on this idea, the useful area of different telescopes, reducers, and cameras was estimated using two different metrics. For the first estimate, regions where the cubic optical distortion component in the direction of the gravitational deflection (i.e., radially outward from the Sun) exceeded the theoretical gravitational deflection in that region were categorized as “not usable.” Thus, if the cubic optical distortion was measured with 1% accuracy, every star position in the “usable” part of the image would be accurate to better than 1%. For example, if the cubic distortion coefficient was measured to be 1 arcsec with a precision of 1%, the gravitational deflection could be measured with uncertainty of 0.01 arcsec or better. While this particular level of precision is likely difficult to achieve in practice, the goal here is to capture the regions of the image where the corrections are comparable to the expected deflections. For the second estimate, the threshold was relaxed so that the “not usable” region was defined as the region where the cubic optical distortion component in the direction of the gravitational deflection exceeded three times the expected gravitational deflection magnitude.

The distances over which these criteria are satisfied were converted to areas in square degrees for different telescope/camera combinations and plotted in [Figure 18](#). This analysis determined that the Askar FRA500 series telescopes paired with $0.7\times$ reducers have very small cubic distortions and therefore have very large usable sky areas when coupled with a medium frame camera like the ZWO ASI2600 and are therefore are ideally suited for measuring the gravitational deflection of starlight. Since the usable area is expected to be near the center of the image where cubic corrections are small, placing the Sun in one corner of the image means many more “high quality” stars will appear in the final data set, thereby reducing the statistical uncertainties.

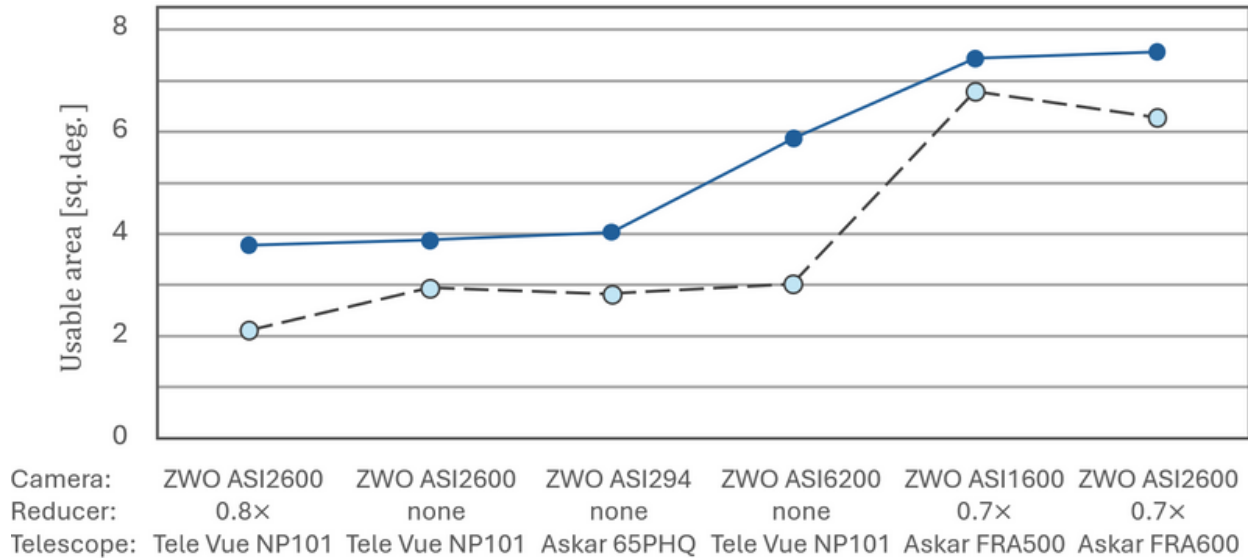


Figure 18

The usable sky area for some telescope and camera combinations is plotted in square degrees. The usable area is the size of the region where the cubic optical distortion C is less than 1.0 (dotted curve) or 3.0 (solid curve) times the theoretical gravitational deflection $\Delta\theta$ in that region. The configurations include the camera model, the focal reducer coefficient (if used), and the telescope model. The data for the Askar FRA600 has not been verified, but the FRA600 is based on the same design as the slightly shorter focal length FRA500 and is therefore expected to have a similar distortion profile. For a given telescope/reducer combination, usable areas should be similar for cameras with similar sensors from other manufacturers.

5.3 Star Field Optimization for MEE2027

The star field for the 2027 eclipse is shown in [Figure 19](#). The Beehive cluster (M44) is in the field of view, which offers many bright stars at a large distance from the Sun. The field of view for the Askar FRA500 with a 0.7× reducer and a ZWO ASI1600MM camera is shown in the upper right corner. The idea is to make a mosaic with the Sun in the corner of each image. Considering that the telescope must slew to these four fields, as well as to two calibration fields 10° away on either side of the Sun, an optimum imaging plan can be formulated. Given that totality in Qena, Egypt (26° 9' 18.4" N, 32° 42' 57.6" E) near the center of the eclipse path will last 370 seconds, we recommend acquiring the two calibration fields during the 30 seconds at the start and end of totality. The four fields with the Sun in the corners should be imaged for 60 seconds each, leaving 70 seconds for mount pointing and settling. Integrated exposures of 30 to 60 seconds for each star field should be enough to significantly reduce daytime atmospheric turbulence, which is expected to be significant at this geographic location. Teams with multiple telescopes might consider dedicating different telescopes to different parts of the mosaic to minimize time spent slewing and waiting for the mount to settle.

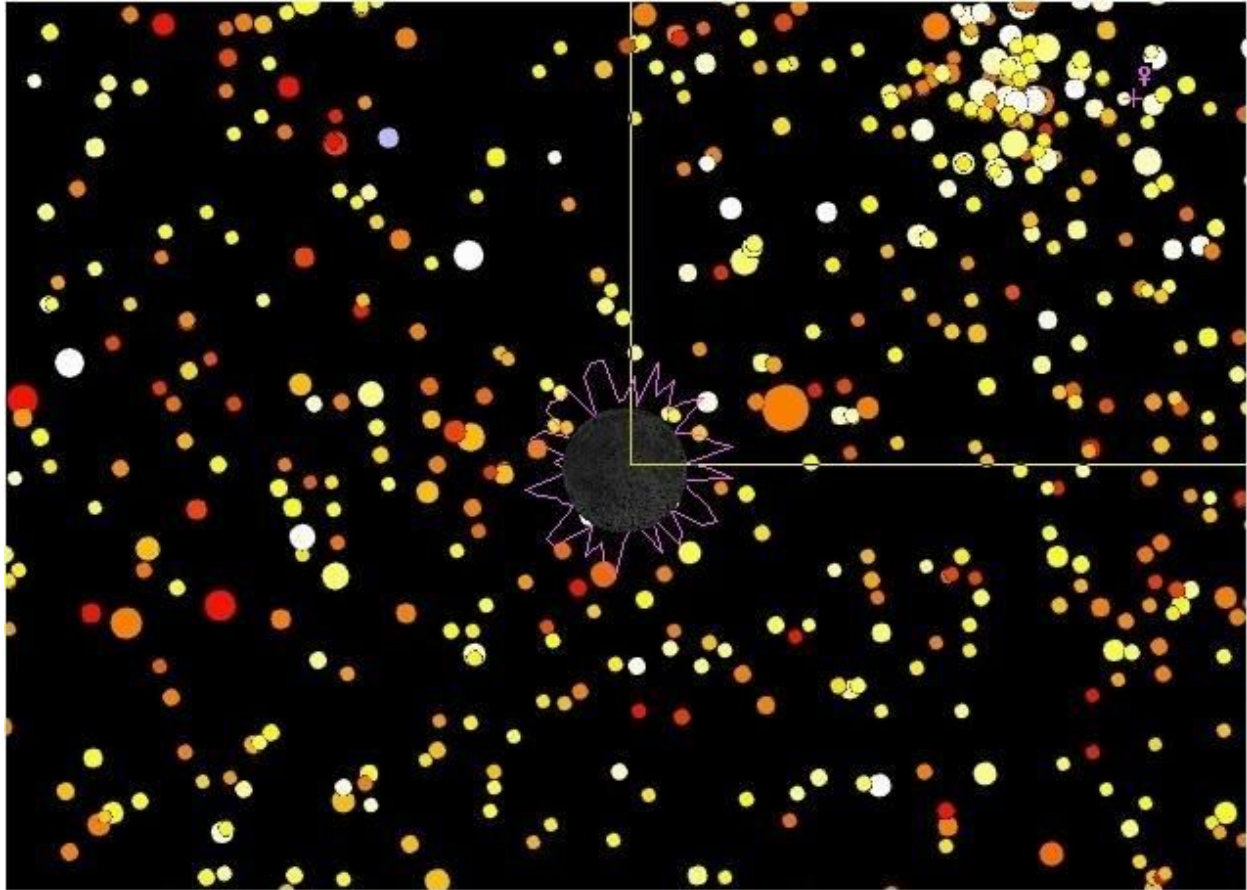


Figure 19

The star field for the 2027 eclipse is shown with stars down to magnitude 11. The colors are representative of the star's spectral class and the dot size indicates the star's brightness with brighter stars indicated by larger dots. If a mosaic is made with the Sun in each of four corners, the number of measurable stars in the “usable area” optimized. The highlighted field at the upper right corner corresponds to the field of view of an Askar FRA500 telescope equipped with a $0.7\times$ reducer and a ZWO ASI1600MM camera. The Beehive cluster (M44) is visible at the top right corner.

5.4 Suitability of Future Eclipses for the Eddington Experiment

As we'll now discuss, of the remaining total eclipses in the 21st century, the eclipses in Northern Africa in 2027 and in Australia in 2028 are best suited for repeating the Eddington experiment (although there will be thirteen additional eclipses over the next 100 or so years during which measuring L may be possible). In order to assess the relative feasibility of obtaining stellar images in the inner corona, a close examination of the field of stars in these thirteen eclipses and the 2024 eclipse was performed. The results of this analysis follow below.

[Figure 20](#) shows the result of a simple sum of the magnitudes of the deflections of all stars brighter than magnitude 9 with distances from the center of the Sun of less than 2 solar radii, weighted by the sine of the angle of the Sun's position above the horizon during the eclipse, i.e.,

$$\Delta\Theta_1 = \sum_i |\Delta\theta_i| \sin \theta_\odot, \quad (3)$$

where $\Delta\theta_i$ is the deflection given by [Equation 1](#) for the i^{th} star in the region of interest ($R < 2$) and $\theta_\odot$ is the elevation of the Sun during the eclipse. This metric provides an estimate the number of stars that are likely to be measurable in inner part of the corona, while also rewarding imaging conditions that are likely to minimize additional distortion introduced by atmospheric turbulence and atmospheric refraction. We also only consider eclipses with long duration over land since precision measurements from floating platforms are unlikely to be feasible and short duration eclipses limit the total integrated acquisition time. The number of measurable bright stars near the solar disk ranges from 1 (in 2024) to 4 (in 2027), while some eclipses have no bright stars near the Sun.

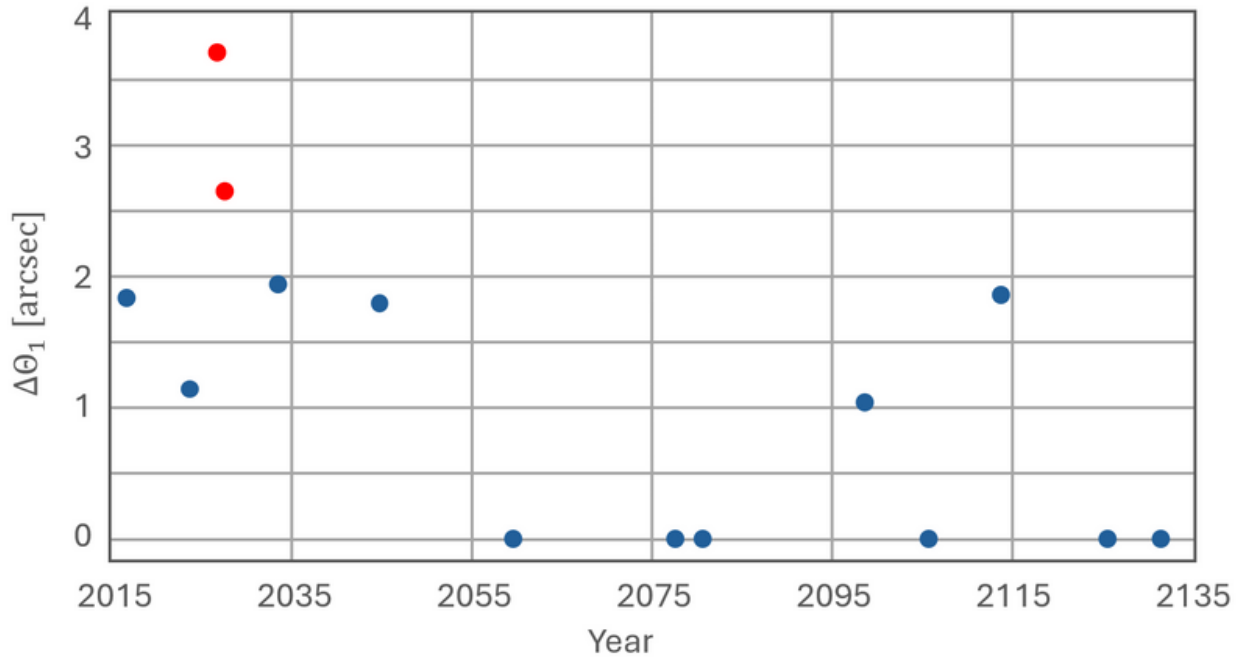


Figure 20

Future eclipses and their relative value $\Delta\Theta_1$ in capturing stars closer than $R = 2$. The horizontal axis indicates the year in which a particular eclipse will occur. The vertical axis is the sum of the magnitudes of the theoretical gravitational deflections of magnitude 9 or lower stars inside $R = 2$ in arcseconds weighted by the sine of the altitude of the Sun's position during totality. As can be seen, by this metric, the eclipses of 2027 and 2028 (highlighted in red) will provide the best opportunity to measure L in the next 100+ years.

[Figure 21](#) shows the geometric sum of the deflections of all stars brighter than magnitude 9 with distance less than 2.5 solar radii from the center of the Sun, weighted by the deflection of a standard star at $R = 2.5$, i.e.,

$$\Delta\Theta_2 = \sum_i \frac{\Delta\theta_i^2}{\Delta\theta_{2.5}}, \quad (4)$$

where $\Delta\theta_i$ is the deflection given by [Equation 1](#) for the i^{th} star in the region of interest ($R < 2.5$) and $\Delta\theta_{2.5}$ is the theoretical deflection of the standard star, i.e., $\Delta\theta(2.5R_\odot) \approx 0.7$ arcsec. This metric weights the stars closest to the Sun more strongly as these are essential to fitting the $1/R$ dependence of $\Delta\theta$, but also rewards being able to image a larger number of stars overall even if these are somewhat further from the disk of the Sun and have more moderate deflections.

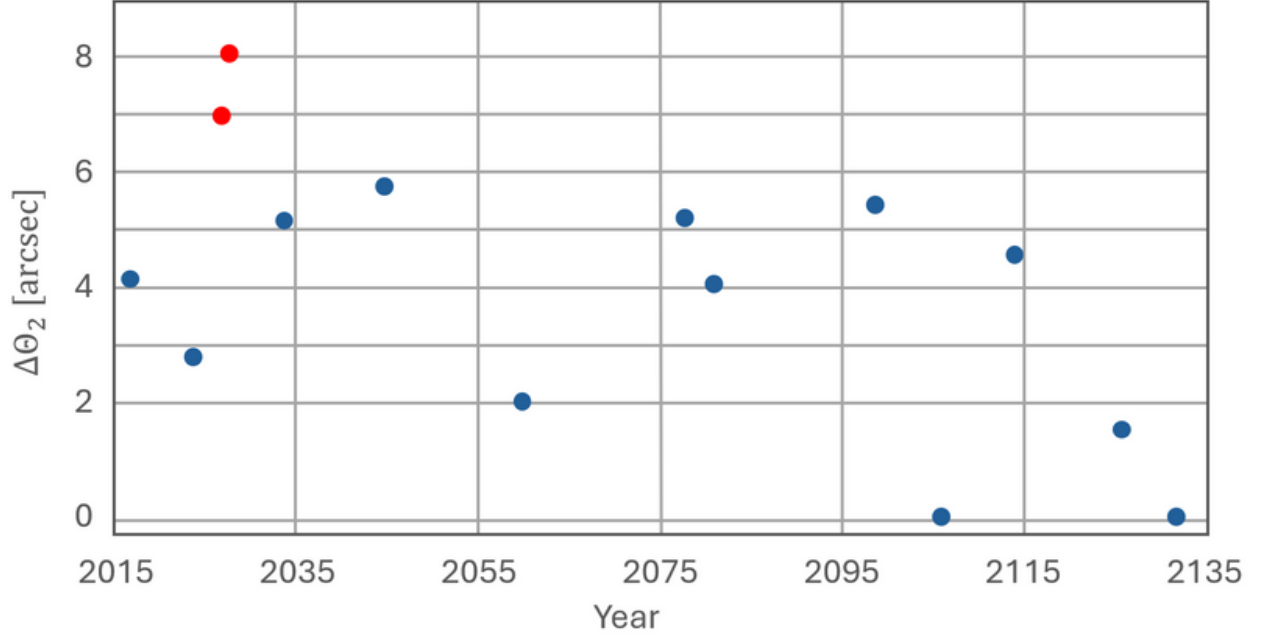


Figure 21

Future eclipses and their relative value $\Delta\Theta_2$ in capturing stars closer than $R = 2.5$. The horizontal axis indicates the year in which a particular eclipse will occur. The vertical axis is the sum of the squares of the theoretical gravitational deflections in arcseconds for stars of magnitude 9 or lower within 2.5 solar radii of the disk of the Sun weighted by the theoretical gravitational deflection of a star at $R = 2.5R_\odot$. As with $\Delta\Theta_1$, this metric suggests that the eclipses of 2027 and 2028 (highlighted in red) provide the best opportunities to measure L in the next 100+ years.

As can be seen from [Figure 20](#) and [Figure 21](#), the eclipses of 2027 and 2028 are ideal for achieving the goal of MEE2024 – creating a curve fit for $\Delta\theta$ as function of R including the deflections of stars in the “forbidden zone” where $R \sim 1$ to 2 solar radii. Together with the discussion in Section 5.3 regarding the number and brightness of stars that will be visible during the 2027 eclipse, we can see that the 2027 eclipse presents the best opportunity for conducting a precision Eddington experiment, followed closely by the eclipse of 2028.

6. Conclusions

The MEE2024 project was mostly successful and most, if not all, of its goals were achieved. First, the effectiveness of modern CMOS cameras to take stellar images near the disk of the Sun was validated.

Essentially continuous imaging could take place since exposure times as low as 100 ms could be used for most of the cameras. This led to the acquisition of 7173 images during the 4.5 minutes of totality. Using these short exposures, also allowed us to image stars closer to the limb of the Sun. The Station 1 data analyzed here and data from other stations at the El Salto site is available for download at <https://www.moderneddingtonexperiment.org>.

Secondly, the number of stations was increased to 13. Even though most of the stations were unable to gather useful data due to the weather, zenith images taken in the nights leading up to the eclipse using a variety of telescopes and cameras allowed for the evaluation of their optical performance. Prior to this analysis, the Tele Vue NP101is was thought to be the ideal telescope because it was known to have very low distortion coefficients (see [Bruns, 2017](#) for a discussion). After evaluation, however, it appears that the Askar FR500 telescope with a $0.7\times$ reducer may actually be the preferred telescope to use in MEE2027 because, surprisingly, it has virtually no distortion near the middle of the field of view and thus a larger usable area.

Next, was the fact that the three teams for the experiment contained students, faculty, and amateur astronomers from several different kinds of institutions. The experience was engaging for all and the professional benefit to the undergraduate students involved in this challenging experiment cannot be overstated. MEE2027 is already attracting participants, including students and faculty from Portugal and Spain, for a practice run during the total eclipse of August 12th, 2026. This eclipse is not particularly useful for accurate determination of L because it is very low to the horizon and thus subject to significant atmospheric distortion, but it will serve well as an opportunity for student involvement and training for MEE2027.

The CMOS cameras all worked well. The ZWO ASI6200MM on Station 1 recorded more than 200 stars with a total of 480 usable frames with exposures of 250, 300 and 400 ms. Despite not slewing away from the eclipse field for calibration, the data collected allowed the determination of Einstein's deflection coefficient as $1.84 \text{ arcsec} \pm 0.24 \text{ arcsec}$. This is consistent with and within 5% of its accepted value of 1.7512 arcsec , although not quite as precise as Bruns' 2017 measurement ([Bruns, 2018](#)). While an attempt was made to use a full frame camera to avoid slewing and capture more stars in a single shot/maximize integration time, it is highly recommended that in the future researchers follow Bruns and slew away from the Sun to acquire calibration fields away from the disk of the Sun. This allows for more precise linear and quadratic corrections, which play an important role in decreasing the uncertainty in the deflection coefficient. The analysis of the cameras used in MEE2024 led to the conclusion that the ZWO ASI2600MM is ideal for MEE2027. This is based on a combination of sensor size, sensor technology/sensitivity, and image download speed.

As far as the location of the sites chosen for MEE2024 is concerned, the three locations should have worked out well, but all were subject to an unusual occurrence of a tropical jet stream storm. It might have been a mistake to place so many telescopes in just three locations since despite a distance of over 850 km between the sites in Mexico and Texas, all were subject to the same weather front. These locations were chosen because on paper they would optimize the duration of the eclipse, altitude of the Sun above the horizon, and the probability

of good weather. The sites in Mexico were also at elevation, which should have helped with atmospheric distortion. However, on the day of the eclipse, locations much further away (even in Vermont and Canada) had much more favorable conditions, despite not being nearly as ideal on paper. The lesson here is that placing telescopes further apart along the eclipse path increases the chance for an organized effort to gather good data. For MEE2027, this might mean having telescopes placed along the northern coast of Africa, as well as near the planned site in Egypt. The eclipse traces a path of over 4000 km across Morocco, Algeria, Tunisia, and Libya, so many options are available, although attention should be paid to historical weather patterns.

Perhaps the greatest success from MEE2024 may prove to be the creation of the open-source MEE data processing software. This allows processing of large data sets created by the CMOS cameras. It not only evaluates the errors introduced by the optics of the imaging system but also finds centroids, stacks frames, calculates gravitational deflections by comparing the centroids from the stacked images to the Gaia catalog, and ultimately computes the deflection coefficient and estimates the uncertainty in the measurement. The software also provides a variety of graphical displays that help understand and optimize the analysis. This integrated software package can be downloaded from GitHub ([Smith & Smith, 2024](#)).

Unfortunately, two goals of the MEE2024 experiment (getting very high precision results and imaging several stars within 1 solar radius of the limb of the Sun) were not realized. Our goal was to obtain an error less than 3% by collecting orders of magnitude more data than was gathered in 2017. Unfortunately, bad weather prevented this. Similarly, no stars closer than 2.5 radii to the Sun were reliably imaged. Despite these two setbacks, we still consider MEE2024 to be an overall success.

As discussed in Section 5.4 above, the total eclipses of 2027 and 2028 are ideal for repeating the Eddington experiment and should allow the measurement of stellar positions in the “forbidden zone”. Other eclipses in the 21st century will not be nearly as well-suited, so if these opportunities are missed, a high precision replication of the Eddington experiment may be postponed for decades. If you have interest in participation in MEE2027 or would like to access the MEE2024 data at www.moderneddingtonexperiment.org, please email Toby Dittrich at tdittrich@pcc.edu or join the conversation at <https://groups.io/g/MEE2027>.

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Footnotes

1. Using the nominal values and uncertainties stipulated for $R_{\odot}$ and for the product $GM_{\odot}$ in IAU 2015 Resolution B3 (Prša et al., 2016) and the definition of the meter in terms of c (Bureau International des Poids et Mesures, 2019). ↵
2. Although this view has recently been challenged (see, e.g., Kennefick (2012) and Schindler (2013)), some historians of science (Earman & Glymour, 1980) have argued that Eddington and Dyson may have been somewhat generous in the interpretation of their data and that they excluded data from the Príncipe site because it did not support Einstein’s prediction. For a comprehensive discussion of this debate, see, Gilmore & Tausch-Pebody (2022). For entertaining accounts of the events leading up to eclipse of 1919, the events of May 29th, and its aftermath, see the excellent books about the Eddington/Dyson expeditions by Gates & Pelletier (2019) and by Kennefick (2021). ↵
3. Note that the absence of attempts at verifying Einstein’s theory using ground-based optical telescopes is likely in part because of the advent of high precision measurements of the gravitational deflection of light made possible by radio telescope arrays (Lebach et al., 1995; Fomalont et al., 2009) and by space-based telescopes (Froeschlé, Mignard & Arenou, 1997), as well as precision tests of relativity using so-called Shapiro time-delay experiments (Shapiro, 1964; Reasenberg et al., 1979; Bertotti, Iess & Tortora, 2003). For a detailed summary of the experimental evidence in support of the theory of general relativity, see C.M. Will’s extensive review (2014). ↵
4. This is useful for developing/practicing eclipse day procedures and testing different imaging configurations/parameters since the brightness of the full Moon is comparable to that of the corona during a total solar eclipse. Naturally, due to the much smaller mass of the Moon any measurements of L taken during the full Moon should be consistent with 0 arcsec. ↵

5.

In a system with no distortion or other source of measurement error, the discrepancy between the measured positions of stars and their catalog positions would correspond exactly to their gravitational deflections.

[↵](#)

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